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June 14, 2023

Elizabeth Mauro
Director, Global Regulatory Science
Pfizer Inc.
66 Hudson Boulevard East
New York, CA 10001

Device: Lucira COVID-19 All-In-One Test Kit

EUA Number: EUA202920

Company: Pfizer Inc.

Indication: Intended to detect the novel coronavirus SARS-CoV-2 that causes COVID-19. This test is for prescription home use with self-collected anterior nasal swab samples from individuals aged 14 years and older who are suspected of COVID-19 by their healthcare provider.

This test is also authorized for use at the point of care (POC), in patient care settings operating under a CLIA Certificate of Waiver, Certificate of Compliance, or Certificate of Accreditation, with self-collected anterior nasal swab samples from individuals aged 14 years and older, and in individuals aged 13 years and under when the specimen is collected by a healthcare provider at the POC.

This test utilizes a molecular amplification technology for the detection of SARS-CoV-2 RNA in individuals with known or suspected COVID-19.

Dear Elizabeth Mauro:

On November 17, 2020, based on Lucira Health, Inc.'s request, the Food and Drug Administration (FDA) issued an Emergency Use Authorization (EUA) for the Lucira COVID-19 All-In-One Test Kit, pursuant to Section 564 of the Federal Food, Drug, and Cosmetic Act (the Act) (21 U.S.C. §360bbb-3) for the indication stated in the letter.¹ In addition, FDA

¹ The November 17, 2020, letter authorized the Lucira COVID-19 All-In-One Test Kit to detect the novel coronavirus SARS-CoV-2 that causes COVID-19. This test was authorized for prescription home use with self-collected nasal swab specimens in individuals aged 14 and older who are suspected of COVID-19 by their healthcare provider. This test was also authorized for use at the point of care (POC), in patient care settings operating under a

established additional Conditions of Authorization in response to the continued emergence of new variants of SARS-CoV-2 on September 23, 2021.² Subsequently, based on Lucira Health, Inc.'s requests, FDA also granted updates to the authorized labeling on July 28, 2022³ and revised and reissued the letter on November 15, 2022.⁴

On April 19, 2023, and May 12, 2023, you⁵ requested that FDA amend your EUA. Based on those requests, and having concluded that revising the November 15, 2022, EUA is appropriate to protect the public health or safety under section 564(g)(2)(C) of the Act (21 U.S.C. § 360bbb-3(g)(2)(C)), FDA is reissuing the November 15, 2022, letter in its entirety with the revisions incorporated.⁶ Pursuant to section 564 of the Act and the Scope of Authorization (Section II) and Conditions of Authorization (Section IV) of this reissued letter, your product⁷ is now authorized for use consistent with the indication described above.

On February 4, 2020, as amended on March 15, 2023, pursuant to Section 564(b)(1)(C) of the Act, the Secretary of the Department of Health and Human Services (HHS) determined that there is a public health emergency, or a significant potential for a public health emergency, that affects, or that has a significant potential to affect, national security or the health and security of United States citizens living abroad, and that involves the virus that causes COVID-19. Pursuant to Section 564 of the Act, and on the basis of such determination, the Secretary of HHS then declared that circumstances exist justifying the authorization of emergency use of in

CLIA Certificate of Waiver, Certificate of Compliance, or Certificate of Accreditation, with self-collected nasal swab specimens in individuals aged 14 and older, and in individuals aged 13 and under when the specimen is collected by a healthcare provider at the POC. This test utilizes molecular amplification technology for the detection of SARS-CoV-2 RNA in individuals with known or suspected COVID-19.

² The Viral Mutation Revision Letter – September 23, 2021, can be accessed at:

<https://www.fda.gov/media/152406/download>

³ On July 28, 2022, Lucira Health, Inc.'s request was granted to update the device shelf-life stability claim from 12 months to 18 months when stored at 30±2°C. FDA also updated the Fact Sheet for Healthcare Providers to reflect language used in more recent authorizations.

⁴ On November 15, 2022, the revisions to the November 17, 2020, letter and authorized labeling included: (1) addition of the following statement in the intended use “*Negative results are presumptive and confirmation with a molecular assay performed in a laboratory, if necessary for patient management, may be performed.*” (2) incorporating language to address Condition of Authorization (1) in the Viral Mutation Revision Letter dated September 23, 2021, (3) updating the titles of some of the authorized labeling, (4) deleting Condition of Authorization P. from the November 17, 2020 letter (fulfilled), (5) deleting Condition of Authorization Q. from the November 17, 2020 letter (fulfilled), and (6) updates to the letter for consistency with language used in more recent authorizations, including incorporation of Conditions of Authorization (2) and (3) from the Viral Mutation Revision Letter – September 23, 2021 (Conditions of Authorization P and Q in the November 15, 2022 letter and also below).

⁵ For ease of reference, this letter will use the term “you” and related terms to refer to Pfizer Inc.

⁶ The revisions to the November 15, 2022, letter and authorized labeling include: (1) transferring ownership of the EUA for the Lucira COVID-19 All-In-One Test Kit from Lucira Health, Inc. to Pfizer Inc., (2) updating the intended use for consistency with language use in other home-use tests, (3) updating the list of authorized distributors, (4) removing the “Lucira Health, Inc. Healthcare Provider Guide to COVID-19 Data Reporting for Non-Lab-Based Testing” from the list of authorized labeling list, instead it will be offered as additional information under Condition of Authorization G. (below), and (5) updating the Fact Sheet for Healthcare Providers and the Letter of Authorization to reflect the updates made to the intended use and for consistency with language used in more recent authorizations.

⁷ For ease of reference, this letter will use the term “your product” to refer to the Lucira COVID-19 All-In-One Test Kit for the indication identified above.

vitro diagnostics for detection and/or diagnosis of the virus that causes COVID-19 subject to the terms of any authorization issued under Section 564(a) of the Act.⁸

FDA considered the totality of scientific information available in authorizing the emergency use of your product for the indication above. A summary of the performance information FDA relied upon is included in the Instructions for Use (identified below).

Having concluded that the criteria for issuance of this authorization under Section 564(c) of the Act are met, I am authorizing the emergency use of your product, described in the Scope of Authorization of this letter (Section II), subject to the terms of this authorization.

I. Criteria for Issuance of Authorization

I have concluded that the emergency use of your product meets the criteria for issuance of an authorization under Section 564(c) of the Act, because I have concluded that:

1. The SARS-CoV-2 can cause a serious or life-threatening disease or condition, including severe respiratory illness, to humans infected by the virus;
2. Based on the totality of scientific evidence available to FDA, it is reasonable to believe that your product may be effective in diagnosing COVID-19, and that the known and potential benefits of your product when used for diagnosing COVID-19, outweigh the known and potential risks of your product; and
3. There is no adequate, approved, and available alternative to the emergency use of your product.⁹

II. Scope of Authorization

I have concluded, pursuant to Section 564(d)(1) of the Act, that the scope of this authorization is limited to the indication above.

Authorized Product Details

Your product is a single-use test kit intended to detect the novel coronavirus SARS-CoV-2 that causes COVID-19. This test is authorized for prescription home use with self-collected anterior nasal swab samples from individuals aged 14 years and older who are suspected of COVID-19 by their healthcare provider. This test is also authorized for use at the point of care (POC), in patient care settings operating under a CLIA Certificate of Waiver, Certificate of Compliance, or

⁸ U.S. Department of Health and Human Services, *Determination of a Public Health Emergency and Declaration that Circumstances Exist Justifying Authorizations Pursuant to Section 564(b) of the Federal Food, Drug, and Cosmetic Act*, 21 U.S.C. § 360bbb-3. February 4, 2020. 85 FR 7316 (February 7, 2020). U.S. Department of Health and Human Services, *Amended Determination of a Public Health Emergency or Significant Potential for a Public Health Emergency Pursuant to Section 564(b) of the Federal Food, Drug, and Cosmetic Act*, 21 U.S.C. § 360bbb-3(b). March 15, 2023. 88 FR 16644 (March 20, 2023) (“Amended Determination”).

⁹ No other criteria of issuance have been prescribed by regulation under Section 564(c)(4) of the Act.

Certificate of Accreditation, with self-collected anterior nasal swab samples from individuals aged 14 years and older, and in individuals aged 13 years and under when the specimen is collected by a healthcare provider at the POC. This test utilizes a molecular amplification technology for the detection of SARS-CoV-2 RNA in patients with known or suspected COVID-19.

SARS-CoV-2 viral RNA is generally detectable in anterior nasal swab samples during the acute phase of infection. Positive results indicate the presence of SARS-CoV-2 viral RNA, but clinical correlation with past medical history and other diagnostic information is necessary to determine patient infection status. Positive results do not rule out bacterial infection or co-infection with other viruses. The agent detected may not be the definitive cause of disease. Individuals who test positive should self-isolate and seek additional care from their healthcare provider.

Negative results are presumptive and confirmation with a molecular assay performed in a laboratory, if necessary for patient management, may be performed. Negative results do not preclude SARS-CoV-2 infection and should not be used as the sole basis for treatment or management decisions for the individual, including infection control decisions. Negative results should be considered in the context of an individual's recent exposures, history and the presence of clinical signs and symptoms consistent with COVID-19. Individuals who test negative and continue to experience COVID-19 like symptoms of fever, cough, and/or shortness of breath may still have SARS-CoV-2 infection and should seek follow up care from their physician or healthcare provider.

All prescribing healthcare providers will report all test results they receive from individuals who use the authorized product to relevant public health authorities in accordance with local, state, and federal requirements, using appropriate LOINC and SNOMED codes, as defined by the Laboratory In Vitro Diagnostics (LIVD) Test Code Mapping for SARS-CoV-2 Tests provided by the Centers for Disease Control and Prevention (CDC).

Your product is performed using self-collected anterior nasal swab samples from individuals aged 14 years and older, and in individuals aged 13 years and under when the specimen is collected by a healthcare provider at the POC. When using your product, the individual performing the test must follow the instructions provided in the "Lucira COVID-19 All-In-One Test Kit" Package Insert when collecting the specimen, running the test procedure and interpreting the results.

The Lucira COVID-19 All-In-One Test Kit includes the materials or other authorized materials (as may be requested under Condition K. below) required to collect the anterior nasal swab sample and perform the test procedure, as described in the "Lucira COVID-19 All-In-One Test Kit" Package Insert and the "Lucira COVID-19 All-In-One Test Kit Instructions for Use."

Your product requires the following control materials, or other authorized control materials (as may be requested under Condition K. below), that are described in the Instructions for Use. Your product also requires the use of additional authorized materials and authorized ancillary reagents that are not included with your product and are described in the Instructions for Use.

The labeling entitled "Lucira Health COVID-19 All-In-One Test Kit" Package Insert, the "Lucira COVID-19 All-In-One Test Kit Instructions for Use"(available at <https://www.fda.gov/medical-devices/coronavirus-disease-2019-covid-19-emergency-use->

[authorizations-medical-devices/in-vitro-diagnostics-euas](#)), and the following fact sheet pertaining to the emergency use, which are required to be made available as set forth in the Conditions of Authorization (Section IV), are collectively referred to as “authorized labeling:”

- Fact Sheet for Healthcare Providers¹⁰: Pfizer Inc.- Lucira COVID-19 All-In-One Test Kit

The above described product, when accompanied by the authorized labeling as set forth in the Conditions of Authorization (Section IV), is authorized to be distributed and used under this EUA, despite the fact that it does not meet certain requirements otherwise required by applicable federal law.

I have concluded, pursuant to Section 564(d)(2) of the Act, that it is reasonable to believe that the known and potential benefits of your product, when used consistent with the Scope of Authorization of this letter (Section II), outweigh the known and potential risks of your product.

I have concluded, pursuant to Section 564(d)(3) of the Act, based on the totality of scientific evidence available to FDA, that it is reasonable to believe that your product may be effective in diagnosing COVID-19, when used consistent with the Scope of Authorization of this letter (Section II), pursuant to Section 564(c)(2)(A) of the Act.

FDA has reviewed the scientific information available to FDA, including the information supporting the conclusions described in Section I above, and concludes that your product (as described in the Scope of Authorization of this letter (Section II)) meets the criteria set forth in Section 564(c) of the Act concerning safety and potential effectiveness.

The emergency use of your product under this EUA must be consistent with, and may not exceed, the terms of this letter, including the Scope of Authorization (Section II) and the Conditions of Authorization (Section IV). Subject to the terms of this EUA and under the circumstances set forth in the Secretary of HHS’s determination under Section 564(b)(1)(C) of the Act described above and the Secretary of HHS’s corresponding declaration under Section 564(b)(1) of the Act, your product is authorized for the indication above.

III. Waiver of Certain Requirements

I am waiving the following requirements for your product during the duration of this EUA:

- Current good manufacturing practice requirements, including the quality system requirements under 21 CFR Part 820 with respect to the design, manufacture, packaging, labeling, storage, and distribution of your product, but excluding Subpart H (Acceptance Activities, 21 CFR 820.80 and 21 CFR 820.86), Subpart I (Nonconforming Product, 21 CFR 820.90), and Subpart O (Statistical Techniques, 21 CFR 820.250).

¹⁰ Note that the information typically found in a Fact Sheet for Patients is contained in the “Lucira Health COVID-19 All-In-One Test Kit” Package Insert.

IV. Conditions of Authorization

Pursuant to Section 564(e) of the Act, I am establishing the following conditions on this authorization:

Pfizer Inc. (You) and Authorized Distributor(s)¹¹

- A. Your product must comply with the following labeling requirements pursuant to FDA regulations: the intended use statement in 21 CFR 809.10(a)(2), (b)(2); adequate directions for use in 21 U.S.C. 352(f) and 21 CFR 809.10(b)(5), (7), and (8); appropriate limitations on the use of the device including information required under 21 CFR 809.10(a)(4); and any available information regarding performance of the device, including requirements under 21 CFR 809.10(b)(12).
- B. You and authorized distributor(s) must make available on your website(s) all authorized labeling identified in Section II of this letter.
- C. You and authorized distributor(s) must maintain records of customer complaint files and report to FDA any significant complaints about usability or deviations from the established performance characteristics of which you and authorized distributor(s) become aware.
- D. You and authorized distributor(s) must inform relevant public health authorities of this EUA, including the terms and conditions herein, and any updates made to your product and/or the authorized labeling.
- E. Through a process of inventory control, you and authorized distributor(s) must maintain records of the locations (e.g., pharmacies, doctor's offices, etc.) to which your product is distributed and the number of your product distributed to each location.
- F. You and authorized distributor(s) must collect information on the performance of your product and have a process in place to track adverse events, including any occurrence of false positive or false negative results and significant deviations from the established performance characteristics of the product of which you become aware and report any such events to FDA in accordance with 21 CFR Part 803. Serious adverse events, especially unexpected biosafety concerns, should immediately be reported to the Division of Microbiology (DMD)/Office of Health Technology 7 (OHT7) /Office of Product Evaluation and Quality (OPEQ)/Center for Devices and Radiological Health (CDRH) (via email: CDRH-EUAREporting@fda.hhs.gov).
- G. You and authorized distributor(s) are authorized to make available additional information relating to the emergency use of your product that is consistent with, and

¹¹ "Authorized Distributor(s)" are identified by you, Pfizer Inc., in your EUA submission as an entity allowed to distribute your product.

does not exceed, the terms of this letter of authorization.

Pfizer Inc. (You)

- H. You must notify FDA of any authorized distributor(s) of your product, including the name, address, and phone number of any authorized distributor(s).
- I. You must provide authorized distributor(s) with a copy of this EUA and communicate to authorized distributor(s) any subsequent revisions that might be made to this EUA and its authorized accompanying materials, including the authorized labeling.
- J. You must include a physical copy of the “Lucira Health COVID-19 All-In-One Test Kit” Package Insert with each shipped kit of your product, and must make the authorized “Lucira COVID-19 All-In-One Test Kit Instructions for Use” electronically available for healthcare providers. Additionally, you must give healthcare providers the opportunity to request a copy of the “Lucira COVID-19 All-In-One Test Kit Instructions for Use” in paper form, and, after such request, promptly provide the requested labeling without additional cost.
- K. You may request changes to this EUA for your product, including to the Scope of Authorization (Section II in this letter) or to the authorized labeling, including requests to make available additional authorized labeling specific to an authorized distributor. Such additional labeling may use another name for the product but otherwise must be consistent with the authorized labeling, and shall not exceed the terms of authorization of this letter. Any request for changes to this EUA should be submitted to DMD/OHT7/OPEQ/CDRH and require appropriate authorization from FDA prior to implementation.
- L. You must comply with the following requirements pursuant to FDA regulations: Subpart H (Acceptance Activities, 21 CFR 820.80 and 21 CFR 820.86), Subpart I (Nonconforming Product, 21 CFR 820.90), and Subpart O (Statistical Techniques, 21 CFR 820.250).
- M. You must have lot release procedures and the lot release procedures, including the study design and statistical power, must ensure that the product released for distribution meet the clinical and analytical performance claimed in the authorized labeling.
- N. If requested by FDA, you must submit your lot release procedures to FDA, including sampling protocols, testing protocols, and acceptance criteria, that you use to release lots of your product for distribution in the U.S. If such lot release procedures are requested by FDA, you must provide them within 48 hours of the request.
- O. You must evaluate the analytical limit of detection and assess traceability¹² of your product with any FDA-recommended reference material(s) if requested by FDA. After

¹² Traceability refers to tracing analytical sensitivity/reactivity back to an FDA-recommended reference material.

submission to and concurrence with the data by FDA, you will update your labeling to reflect the additional testing if requested by FDA. Such labeling updates will be made in consultation with, and require concurrence of, DMD/OHT7/OPEQ/CDRH.

- P. You must evaluate the impact of SARS-CoV-2 viral mutations on your product's performance. Such evaluations must occur on an ongoing basis and must include any additional data analysis that is requested by FDA in response to any performance concerns you or FDA identify during routine evaluation. Additionally, if requested by FDA, you must submit records of these evaluations for FDA review within 48 hours of the request. If your evaluation identifies viral mutations that affect the stated expected performance of your device, you must notify FDA immediately (via email: CDRH-EUA-Reporting@fda.hhs.gov).
- Q. If requested by FDA, you must update your labeling within 7 calendar days to include any additional labeling risk mitigations identified by FDA regarding the impact of viral mutations on test performance. Such updates will be made in consultation with, and require concurrence of, DMD/OHT7/OPEQ/CDRH.

Healthcare Providers

- R. All prescribing healthcare providers must collect information on the performance of your product in the ordinary course of business and report to DMD/OHT7/OPEQ/CDRH (via email: CDRH-EUA-Reporting@fda.hhs.gov) and to you (1-888-582-4724) any suspected occurrence of false positive or false negative results and significant deviations from the established performance characteristics of your product of which they become aware.
- S. All prescribing healthcare providers must report all test results they receive from individuals who use your product to relevant public health authorities in accordance with local, state, and federal requirements, using appropriate LOINC and SNOMED codes, as defined by the Laboratory In Vitro Diagnostics (LIVD) Test Code Mapping for SARS-CoV-2 Tests provided by the Centers for Disease Control (available at: <https://www.cdc.gov/csels/dls/sars-cov-2-livd-codes.html>). Healthcare providers will also report to Lucira Health, Inc., when requested by Lucira Health, Inc., how many individuals reported test results compared to how many of your product they prescribed.

Point-of-Care Settings¹³

- T. Point-of-care settings using your product must use your product without any deviations from the authorized labeling.
- U. Point-of-care settings that receive your product must notify the relevant public health authorities of their intent to use your product prior to initiating testing.

¹³ For purposes of this EUA, Point-of-Care (POC) settings are defined as patient care settings operating under a CLIA Certificate of Waiver, Certificate of Compliance, or Certificate of Accreditation.

- V. Point-of-care settings using your product must have a process in place for reporting test results to healthcare providers and relevant public health authorities, as appropriate.
- W. Point-of-care settings must collect information on the performance of your product and report to DMD/OHT7/OPEQ/CDRH (via email: CDRH-EUA-Reporting@fda.hhs.gov) and to you (1-888-582-4724) any suspected occurrence of false positive or false negative results and significant deviations from the established performance characteristics of your product of which they become aware.

Pfizer Inc. (You), Authorized Distributors, Healthcare Providers and Point-of-Care Settings

- X. You, authorized distributors, healthcare providers and point-of-care settings using your product must ensure that any records associated with this EUA are maintained until otherwise notified by FDA. Such records will be made available to FDA for inspection upon request.

Conditions Related to Printed Materials, Advertising and Promotion

- Y. All descriptive printed matter, advertising, and promotional materials, relating to the use of your product shall be consistent with the authorized labeling, as well as the terms set forth in this EUA and meet the requirements set forth in section 502(a), (q)(1), and (r) of the Act, as applicable, and FDA implementing regulations.
- Z. No descriptive printed matter, advertising, or promotional materials, relating to the use of your product may represent or suggest that this test is safe or effective for the detection of SARS-CoV-2.
- AA. All descriptive printed matter, advertising, and promotional materials, relating to the use of your product shall clearly and conspicuously state that:
- This product has not been FDA cleared or approved but has been authorized by FDA for emergency use under an EUA;
 - This product has been authorized only for the detection of nucleic acid from SARS-CoV-2, not for any other viruses or pathogens; and
 - The emergency use of this product is only authorized for the duration of the declaration that circumstances exist justifying the authorization of emergency use of in vitro diagnostics for detection and/or diagnosis of COVID-19 under Section 564(b)(1) of the Federal Food, Drug, and Cosmetic Act, 21 U.S.C. § 360bbb-3(b)(1), unless the declaration is terminated or authorization is revoked sooner.

The emergency use of your product as described in this letter of authorization must comply with the conditions and all other terms of this authorization.

V. Duration of Authorization

This EUA will be effective until the declaration that circumstances exist justifying the authorization of the emergency use of in vitro diagnostics for detection and/or diagnosis of COVID-19 is terminated under Section 564(b)(2) of the Act or the EUA is revoked under Section 564(g) of the Act.

Sincerely,

Jeffrey E. Shuren, M.D., M.P.H.
Director
Center for Devices and Radiological Health
Food and Drug Administration

Enclosure



April 2, 2025

Elizabeth Mauro
Director, Global Regulatory Science
Pfizer Inc.
66 Hudson Boulevard East
New York, NY 10001
Re: Revocation of EUA202920

Dear Elizabeth Mauro:

This letter is in response to the request from Pfizer Inc., in letter dated March 14, 2025, that the U.S. Food and Drug Administration (FDA) revoke the EUA for the Lucira COVID-19 All-In-One Test Kit issued on November 17, 2020, amended on September 23, 2021, and July 28, 2022, and revised and reissued on November 15, 2022, and June 14, 2023. Pfizer Inc. indicated that they did not distribute the authorized product and requested that the EUA be revoked. FDA understands that as of the date of this letter there are no viable Lucira COVID-19 All-In-One Test Kit reagents in distribution in the United States.

The authorization of a device for emergency use under section 564 of the Federal Food, Drug, and Cosmetic Act (the Act) (21 U.S.C. 360bbb-3) may, pursuant to section 564(g)(2) of the Act, be revoked when circumstances make such revocation appropriate to protect the public health or safety (section 564(g)(2)(C) of the Act). Because Pfizer Inc. has requested that FDA revoke the EUA for the Lucira COVID-19 All-In-One Test Kit, FDA has determined that it is appropriate to protect the public health or safety to revoke this authorization. Accordingly, FDA hereby revokes EUA202920 for the Lucira COVID-19 All-In-One Test Kit, pursuant to section 564(g)(2)(C) of the Act. As of the date of this letter, the Lucira COVID-19 All-In-One Test Kit is no longer authorized for emergency use by FDA.

Notice of this revocation will be published in the *Federal Register*, pursuant to section 564(h)(1) of the Act.

Sincerely,

Ellen J. Flannery, J.D.
Deputy Center Director for Policy
Director, Office of Policy
Center for Devices and Radiological Health
Food and Drug Administration



February 16, 2023

Myriam Battistutta
Head of Regulatory
Ellume Limited
57 Didsbury Street
East Brisbane QLD, 4169 Australia

Device:	Ellume COVID-19 Home Test
EUA Number:	EUA203011
Company:	Ellume Limited
Indication:	Non-prescription home use for the qualitative detection of nucleocapsid protein antigen from the SARS-CoV-2 virus with self-collected direct mid-turbinate nasal swab samples from individuals aged 16 years or older, or adult collected direct mid-turbinate nasal swab samples from individuals aged 2 years or older. This test is authorized for individuals with symptoms of COVID-19 within the first 4 days of symptom onset when tested at least twice over three days with at least 48 hours between tests, and for individuals without symptoms or other epidemiological reasons to suspect COVID-19, when tested at least three times over five days with at least 48 hours between tests.

Dear Myriam Battistutta:

On December 15, 2020, based on your¹ request the Food and Drug Administration (FDA) issued an Emergency Use Authorization (EUA) for the Ellume COVID-19 Home Test pursuant to Section 564 of the Federal Food, Drug, and Cosmetic Act (the Act) (21 U.S.C. §360bbb-3) for the indication stated in the letter.² Based on your requests, FDA granted updates to the authorized labeling on February 11, 2021³ and also reissued the letter in its entirety with

¹ For ease of reference, this letter will use the term “you” and related terms to refer to Ellume Limited (“Ellume”).

² The December 15, 2020, letter authorized the Ellume COVID-19 Home Test for the qualitative detection of SARS-CoV-2 nucleocapsid antigens from mid-turbinate nasal swabs that are self-collected by an individual age 16 years or older, or are collected by an adult from an individual 2 years of age and older for use in individuals with or without symptoms or other epidemiological reasons to suspect a COVID-19 infection.

The Ellume COVID-19 Home Test was authorized for non-prescription home use.

³ On February 11, 2021, your request was granted via email to update the authorized labeling to address Condition of Authorization R and S. of the December 15, 2020, letter.

revisions incorporated on March 16, 2022.⁴ In addition, FDA established additional Conditions of Authorization in response to the continued emergence of new variants of SARS-CoV-2 on September 23, 2021.⁵ Further, FDA revised the authorized uses and established one additional Condition of Authorization requiring updates to product labeling regarding repeat, or serial, testing, for all currently authorized SARS-CoV-2 antigen tests on November 1, 2022.⁶

On May 8, 2022, July 13, 2022, and November 15, 2022, you requested to amend your EUA. Based on these requests, and having concluded that revising the March 16, 2022, EUA is appropriate to protect the public health or safety under section 564(g)(2)(C) of the Act (21 U.S.C. § 360bbb-3(g)(2)(C)), FDA is revising and reissuing the March 16, 2022, letter in its entirety with the revisions incorporated.⁷ Pursuant to section 564 of the Act and the Scope of Authorization (Section II) and Conditions of Authorization (Section IV) of this reissued letter, your product⁸ is now authorized for use consistent with the indication described above.

On February 4, 2020, pursuant to Section 564(b)(1)(C) of the Act, the Secretary of the Department of Health and Human Services (HHS) determined that there is a public health emergency that has a significant potential to affect national security or the health and security of

⁴ On March 16, 2022, the revisions to the December 15, 2020, letter and authorized labeling included: (1) revisions to the intended use to: “*self-collected direct mid-turbinate nasal swab specimens from individuals aged 16 years or older with symptoms of COVID-19 within the first 7 days of symptom onset*,” “*adult-collected mid-turbinate nasal swab specimens from individuals aged 2 years or older with symptoms of COVID-19 within the first 7 days of symptom onset*,” and “*with self-collected mid-turbinate nasal swab specimens from individuals aged 16 years or older, or adult-collected mid-turbinate nasal swab specimens from individuals aged 2 years or older, with or without symptoms or other epidemiological reasons to suspect COVID-19 when tested twice over two to three days with at least 24 hours (and no more than 48 hours) between tests*,” (2) updates to the Fact Sheet for Healthcare Providers and Fact Sheet for Patients to reflect the revised intended use and also for consistency with language used in more recent authorizations, (3) updates to the letter to reflect the revised intended use including the addition of Condition of Authorization S. (below), (4) updates to the other authorized labeling to reflect the revised intended use, (5) addition of Condition of Authorization R. (below) for additional clinical evaluation and inclusion of Conditions of Authorization (2) and (3) from the Viral Mutation Revision Letter – September 23, 2021 (T. and U. below), (6) removal of Condition of Authorization R. and S. from the December 15, 2020 letter (fulfilled) and Q. from the December 15, 2020 letter (due to revised intended use), (7) updates to the letter and authorized labeling for consistency with language used in more recent authorizations, and (8) updates to the Ellume COVID-19 Home Test software application user interface. A technical correction to the March 16, 2022, letter was issued March 18, 2022, to correct the description of the controls and control materials.

⁵ The Viral Mutation Revision Letter – September 23, 2021, can be accessed at:

<https://www.fda.gov/media/152406/download>.

⁶ The Repeat Testing Revision Letter - November 1, 2022, can be accessed at:

<https://www.fda.gov/media/162799/download>.

⁷ The revisions to the March 16, 2022 letter and authorized labeling include: (1) incorporating your response to Condition of Authorization (1) of the Repeat Testing Revision Letter dated November 1, 2022 to revise the authorized use(s) as required and described in Appendix A, and make various updates to the authorized labeling as required and described in Appendix B of the letter, (2) revisions to the intended use to “*individuals with symptoms of COVID-19 within the first 4 days of symptom onset*” (previously within the first 7 days of symptom onset), (3) deleting Conditions of Authorization Q. and S. in the March 16, 2022, letter as fulfilled through data and information submitted to the Agency and associated updates to the authorized labeling, (4) deleting Condition of Authorization R. in the March 16, 2022, letter in accordance with the Repeat Testing Revision Letter dated November 1, 2022, (5) added Condition of Authorization S. (below) to be consistent with more recent over-the-counter authorizations, (7) updating the informational video’s from mandatory to optional viewing prior to performing the test, and (8) updating the Letter of Authorization and Fact Sheet for Healthcare Professionals to reflect the revised intended use, and for consistency with language used in more recent authorizations.

⁸ For ease of reference, this letter will use the term “your product” to refer to the Ellume COVID-19 Home Test used for the indication identified above.

United States citizens living abroad, and that involves the virus that causes COVID-19. Pursuant to Section 564 of the Act, and on the basis of such determination, the Secretary of HHS then declared that circumstances exist justifying the authorization of emergency use of in vitro diagnostics for detection and/or diagnosis of the virus that causes COVID-19 subject to the terms of any authorization issued under Section 564(a) of the Act.⁹

FDA considered the totality of scientific information available in authorizing the emergency use of your product for the indication above. A summary of the performance information FDA relied upon is contained in the “Ellume COVID-19 Home Test Product Overview for Healthcare Professionals” (identified below).

Having concluded that the criteria for issuance of this authorization under Section 564(c) of the Act are met, I am authorizing the emergency use of your product (as described in the Scope of Authorization of this letter (Section II)) in certain individuals for the detection of SARS-CoV-2 subject to the terms of this authorization.

I. Criteria for Issuance of Authorization

I have concluded that the emergency use of your product meets the criteria for issuance of an authorization under Section 564(c) of the Act, because I have concluded that:

1. The SARS-CoV-2 can cause a serious or life-threatening disease or condition, including severe respiratory illness, to humans infected by this virus;
2. Based on the totality of scientific evidence available to FDA, it is reasonable to believe that your product may be effective in diagnosing COVID-19, and that the known and potential benefits of your product, when used for diagnosing COVID-19, outweigh the known and potential risks of such product; and,
3. There is no adequate, approved, and available alternative to the emergency use of your product for diagnosing COVID-19.¹⁰

II. Scope of Authorization

I have concluded, pursuant to Section 564(d)(1) of the Act, that the scope of this authorization is limited to the indication above.

The Authorized Product

Your product is a rapid, lateral flow immunoassay intended for the qualitative detection of nucleocapsid protein antigen from the SARS-CoV-2 virus. This test is authorized for non-prescription home use with self-collected direct mid-turbinate nasal swab samples from individuals aged 16 years or older or adult collected mid-turbinate swab samples from individuals aged 2 years or older. This test is authorized for individuals with symptoms of

⁹ U.S. Department of Health and Human Services, *Determination of a Public Health Emergency and Declaration that Circumstances Exist Justifying Authorizations Pursuant to Section 564(b) of the Federal Food, Drug, and Cosmetic Act*, 21 U.S.C. § 360bbb-3. 85 FR 7316 (February 7, 2020).

¹⁰ No other criteria of issuance have been prescribed by regulation under Section 564(c)(4) of the Act.

COVID-19 within the first 4 days of symptom onset when tested at least twice over three days with at least 48 hours between tests and for individuals without symptoms or other epidemiological reasons to suspect COVID-19 when tested at least three times over five days with at least 48 hours between tests. The Ellume COVID-19 Home Test does not differentiate between SARS-CoV and SARS-CoV-2.

The SARS-CoV-2 nucleocapsid protein antigen is generally detectable in mid-turbinate nasal swab specimens during the acute phase of infection. Positive results indicate the presence of viral antigens, but the clinical correlation with past medical history and other diagnostic information is necessary to determine infection status. Positive results do not rule out a bacterial infection or co-infection with other viruses and the agent detected may not be the definitive cause of disease. Individuals who test positive with the Ellume COVID-19 Home Test should self-isolate and seek follow up care with their physician or healthcare provider as additional testing may be necessary.

Negative results are presumptive and confirmation with a molecular assay, if necessary, for patient management may be performed. Negative results do not rule out SARS-CoV-2 infection and should not be used as the sole basis for treatment or patient management decisions, including infection control measures such as isolating from others and wearing masks. Negative results should be considered in the context of an individual's recent exposures, history and the presence of clinical signs and symptoms consistent with COVID-19.

Test results will be reported to relevant public health authorities in accordance with local, state, and federal requirements, using appropriate LOINC and SNOMED codes, as defined by the Laboratory In Vitro Diagnostics (LIVD) Test Code Mapping for SARS-CoV-2 Tests provided by the Centers for Disease Control and Prevention (CDC). Automatic test result reporting occurs via the Ellume COVID-19 Home Test software application (app).

Your product is performed using mid-turbinate nasal swab specimens from individuals aged 2 years and older. When using your product, the individual performing the test must first read the “Ellume COVID-19 Home Test Product Information Leaflet” and the “Ellume COVID-19 Home Test Quick Start Guide” to download the Ellume COVID-19 Home Test App onto a compatible smartphone device,¹¹ and then connect the single use Analyzer to their smartphone. The individual performing the test has the option to view an informational video before they must follow the step-by-step instructions provided in the Ellume COVID-19 Home Test App when collecting the specimen, running the test procedure and interpreting the results provided from the Analyzer through the App.

The Ellume COVID-19 Home Test includes the materials or other authorized materials (as may be requested under Condition K. and L. below), required to collect the mid-turbinate nasal swab specimen and perform the test procedure and detect the result, as described in the “Ellume COVID-19 Home Test Product Overview for Healthcare Professionals,” the “Ellume COVID-19 Home Test Product Information Leaflet” and the “Ellume COVID-19 Home Test Quick Start Guide”.

Your product requires the following internal control materials, or other authorized control

¹¹ The Ellume COVID-19 Home Test must be run using a mobile software application, the Ellume COVID-19 Home Test App, on a compatible smartphone device. Compatible smartphone devices include an ARM64 Apple iPhone device with Bluetooth running iOS 12 or later versions of the OS, and Android Phones with Bluetooth 4.0 or later and XHDPI (720x1280px) resolution supporting Android API 23 (version 6.0) or later. Additional smartphone models as may be requested, and for which you receive appropriate authorization, in accordance with Condition K. below.

materials (as may be requested under Condition K below), that are processed along with the specimens. All internal controls must generate expected results in order for a test result to be considered valid, as outlined in the “Ellume COVID-19 Home Test Product Overview for Healthcare Professionals.”

The following labeling is collectively referred to as “authorized labeling”:

- “Ellume COVID-19 Home Test App” software application
- “COVID-19 Home Test Animated Instructions Video 2-12(Commercial)”
- “COVID-19 Home Test Animated Instructions Video 13+(Commercial)”
- “COVID19 Home Test Result Emails Specification” and,
- the “Ellume COVID-19 Home Test Product Information Leaflet”, “Ellume COVID-19 Home Test” box label¹², “Ellume COVID-19 Home Test – Quick Start Guide”, “Ellume COVID-19 Home Test Product Overview for Healthcare Professionals”, “Ellume COVID-19 Home Test – Frequently Asked Questions,” and the Fact Sheet for Healthcare Professionals¹³, which are available at <https://www.fda.gov/medical-devices/coronavirus-disease-2019-covid-19-emergency-use-authorizations-medical-devices/in-vitro-diagnostics-euas>.

Your product, when accompanied by the authorized labeling, is authorized to be distributed to and used by individuals, as set forth in this EUA, despite the fact that it does not meet certain requirements otherwise required by applicable federal law.

I have concluded, pursuant to Section 564(d)(2) of the Act, that it is reasonable to believe that the known and potential benefits of your product, when used consistent with the Scope of Authorization of this letter (Section II), outweigh the known and potential risks of such product.

I have concluded, pursuant to Section 564(d)(3) of the Act, based on the totality of scientific evidence available to FDA, that it is reasonable to believe that your product may be effective in diagnosing COVID-19 when used consistent with the Scope of Authorization of this letter (Section II), pursuant to Section 564(c)(2)(A) of the Act.

FDA has reviewed the scientific information available to FDA, including the information supporting the conclusions described in Section I above, and concludes that your product, when used consistent with the Scope of Authorization of this letter (Section II)), meets the criteria set forth in Section 564(c) of the Act concerning safety and potential effectiveness.

The emergency use of your product under this EUA must be consistent with, and may not exceed, the terms of this letter, including the Scope of Authorization (Section II) and the Conditions of Authorization (Section IV). Subject to the terms of this EUA and under the circumstances set forth in the Secretary of HHS's determination under Section 564(b)(1)(C) of

¹² The “Ellume COVID-19 Home Test” box label include boxes for 1-test per box, and “Ellume COVID-19 Home Test” box labels for additional test kits numbers/options as may be requested, and for which you receive appropriate authorization, in accordance with Condition L. below. Ellume COVID-19 Home Test kits numbers/options are described in the “Ellume COVID-19 Home Test Product Overview for Healthcare Professionals” and the “Ellume COVID-19 Home Test Product Information Leaflet.”

¹³ Note that the information typically found in a Fact Sheet for Patients is contained in the authorized “Ellume COVID-19 Home Test - Frequently Asked Questions,” that will be available to end users as set forth in the Conditions of Authorization (Section IV).

the Act described above and the Secretary of HHS’s corresponding declaration under Section 564(b)(1) of the Act, your product is authorized for the indication above.

III. Waiver of Certain Requirements

I am waiving the following requirements for your product for the duration of this EUA:

- Current good manufacturing practice requirements, including the quality system requirements under 21 CFR Part 820 with respect to the design, manufacture, packaging, labeling, storage, and distribution of your product, but excluding Subpart H (Acceptance Activities, 21 CFR 820.80 and 21 CFR 820.86), Subpart I (Nonconforming Product, 21 CFR 820.90), and Subpart O (Statistical Techniques, 21 CFR 820.250).

IV. Conditions of Authorization

Pursuant to Section 564(e) of the Act, I am establishing the following conditions on this authorization:

Ellume Limited (You) and Authorized Distributor(s) ¹⁴

- A. Your product must comply with the following labeling requirements pursuant to FDA regulations: the intended use statement (21 CFR 809.10(a)(2), (b)(2)); adequate directions for use (21 U.S.C. 352(f)), (21 CFR 809.10(b)(5), (7), and (8)); appropriate limitations on the use of the device including information required under 21 CFR 809.10(a)(4); and any available information regarding performance of the device, including requirements under 21 CFR 809.10(b)(12).
- B. You and authorized distributor(s) must inform relevant public health authorities of this EUA, including the terms and conditions herein, and any updates made to your product and/or the authorized labeling.
- C. Through a process of inventory control, you and authorized distributor(s) must maintain records of the locations (e.g., pharmacies, doctor’s offices, etc.) to which your product is distributed and the number of your product distributed to each location.
- D. You and authorized distributor(s) must maintain records of customer complaint files and report to FDA any significant complaints about usability or deviations from the established performance characteristics of the product of which you and authorized distributor(s) become aware.
- E. You and authorized distributor(s) must collect information on the performance of your product and have a process in place to track adverse events, including any occurrence of false positive or false negative results and significant deviations from the established performance characteristics of the product of which you become aware and report any

¹⁴ “Authorized Distributor(s)” are identified by you, Ellume Limited, in your EUA submission as an entity allowed to distribute your product.

such events to FDA in accordance with 21 CFR Part 803. Serious adverse events, especially unexpected biosafety concerns, should immediately be reported to the Division of Microbiology (DMD)/Office of Health Technology 7 (OHT7)-Office of In Vitro Diagnostics/Office of Product Evaluation and Quality (OPEQ)/Center for Devices and Radiological Health (CDRH) (via email: CDRH-EUAREporting@fda.hhs.gov).

- F. You and authorized distributor(s) are authorized to make available additional information relating to the emergency use of your product that is consistent with, and does not exceed, the terms of this letter of authorization.
- G. You and authorized distributor(s) using your product must ensure that any records associated with this EUA are maintained until otherwise notified by FDA. Such records will be made available to FDA for inspection upon request.

Ellume Limited (You)

- H. You must notify FDA of any authorized distributor(s) of your product, including the name, address, and phone number of any authorized distributor(s).
- I. You must provide authorized distributor(s) with a copy of this EUA and communicate to authorized distributor(s) any subsequent revisions that might be made to this EUA and its authorized accompanying materials, including the authorized labeling.
- J. You must make the authorized “Ellume COVID-19 Home Test Product Overview for Healthcare Professionals” (HCP IFU), “Ellume COVID-19 Home Test – Frequently Asked Questions,” and the Fact Sheet for Healthcare Professionals electronically available on your website. Additionally, you must provide the opportunity to request a copy of the HCP IFU and Fact Sheet for Healthcare Professionals in paper form, and after such request, promptly provide the requested labeling at no additional cost.
- K. You may request changes to this EUA for your product, including to the Scope of Authorization (Section II in this letter) or to the authorized labeling, including requests to make available additional authorized labeling specific to an authorized distributor. Such additional labeling may use another name for the product but otherwise must be consistent with the authorized labeling and shall not exceed the terms of authorization of this letter. Any request for changes to this EUA should be submitted to DMD/OHT7/OPEQ/CDRH and require appropriate authorization from FDA prior to implementation.
- L. You may request new box labels to allow additional test kits numbers/options for your product. Such additional labeling requests to this EUA should be submitted to and require concurrence of DMD/OHT7/OPEQ/CDRH prior to implementation.
- M. You will comply with the following requirements pursuant to FDA regulations: 21 CFR Part 820 Subpart H (Acceptance Activities, 21 CFR 820.80 and 21 CFR 820.86), Subpart I (Nonconforming Product, 21 CFR 820.90), and Subpart O (Statistical Techniques, 21 CFR 820.250).

- N. You must have lot release procedures and the lot release procedures, including the study design and statistical power, must ensure that the tests released for distribution have the clinical and analytical performance claimed in the authorized labeling.
- O. If requested by FDA, you must submit lot release procedures to FDA, including sampling protocols, testing protocols, and acceptance criteria, that you use to release lots of your product for distribution in the U.S. If such lot release procedures are requested by FDA, you must provide it within 48 hours of the request.
- P. You must evaluate the analytical limit of detection and assess traceability¹⁵ of your product with any FDA-recommended reference material(s). After submission to and concurrence with the data by FDA, you will update your labeling to reflect the additional testing. Such labeling updates will be made in consultation with, and require concurrence of, DMD/OHT7/OPEQ/CDRH.
- Q. You must evaluate the impact of SARS-CoV-2 viral mutations on your product's performance. Such evaluations must occur on an ongoing basis and must include any additional data analysis that is requested by FDA in response to any performance concerns you or FDA identify during routine evaluation. Additionally, if requested by FDA, you must submit records of these evaluations for FDA review within 48 hours of the request. If your evaluation identifies viral mutations that affect the stated expected performance of your device, you must notify FDA immediately (via email: CDRHEUA-Reporting@fda.hhs.gov).
- R. If requested by FDA, you must update your labeling within 7 calendar days to include any additional labeling risk mitigations identified by FDA, such as those related to the impact of viral mutations on test performance. Such updates will be made in consultation with, and require concurrence of, DMD/OHT7/OPEQ/CDRH.
- S. You must submit your product for any FDA-recommended independent evaluation to confirm the performance characteristics of your test, if requested by FDA. After submission to and concurrence with the data by FDA, you will update your labeling to reflect the additional testing. Such labeling updates will be made in consultation with, and require concurrence of, DMD/OHT7/OPEQ/CDRH.

Conditions Related to Printed Materials, Advertising and Promotion

- T. All descriptive printed matter, advertising and promotional materials relating to the use of your product shall be consistent with the authorized labeling, as well as the terms set forth in this EUA and meet the requirements set forth in section 502(a), (q)(1), and (r) of the Act, as applicable, and FDA implementing regulations.
- U. No descriptive printed matter, advertising or promotional materials relating to the use of your product may represent or suggest that this test is safe or effective for the detection of SARS-CoV-2.

¹⁵ Traceability refers to tracing analytical sensitivity/reactivity back to an FDA-recommended reference material.

V. All descriptive printed matter, advertising and promotional materials relating to the use of your product shall clearly and conspicuously state that:

- This product has not been FDA cleared or approved; but has been authorized by FDA under an EUA;
- This product has been authorized only for the detection of proteins from SARS- CoV-2, not for any other viruses or pathogens; and,
- The emergency use of this product is only authorized for the duration of the declaration that circumstances exist justifying the authorization of emergency use of in vitro diagnostics for detection and/or diagnosis of COVID-19 under Section 564(b)(1) of the Federal Food, Drug and Cosmetic Act, 21 U.S.C. § 360bbb-3(b)(1), unless the declaration is terminated or authorization is revoked sooner.

The emergency use of your product as described in this letter of authorization must comply with the conditions and all other terms of this authorization.

V. Duration of Authorization

This EUA will be effective until the declaration that circumstances exist justifying the authorization of the emergency use of in vitro diagnostics for detection and/or diagnosis of COVID-19 is terminated under Section 564(b)(2) of the Act or the EUA is revoked under Section 564(g) of the Act.

Sincerely,

Jeffrey E. Shuren, M.D., J.D.
Director
Center for Devices and Radiological Health
Food and Drug Administration

FDA NEWS RELEASE

Coronavirus (COVID-19) Update: FDA Authorizes Antigen Test as First Over-the-Counter Fully At-Home Diagnostic Test for COVID-19

Test is authorized for individuals two years of age or older, including those not showing symptoms

[More Press Announcements \(/news-events/newsroom/press-announcements\)](/news-events/newsroom/press-announcements)

For Immediate Release:

December 15, 2020

[Español \(/node/395724\)](/node/395724)

Today, the U.S. Food and Drug Administration issued an emergency use authorization (EUA) for the first over-the-counter (OTC) fully at-home diagnostic test for COVID-19. The [Ellume COVID-19 Home Test \(/media/144457/download\)](/media/144457/download) is a rapid, lateral flow antigen test, a type of test that runs a liquid sample along a surface with reactive molecules. The test detects fragments of proteins of the SARS-CoV-2 virus from a nasal swab sample from any individual 2 years of age or older.

“Today’s authorization is a major milestone in diagnostic testing for COVID-19. By authorizing a test for over-the-counter use, the FDA allows it to be sold in places like drug stores, where a patient can buy it, swab their nose, run the test and find out their results in as little as 20 minutes,” said FDA Commissioner Stephen M. Hahn, M.D. “As we continue to authorize additional tests for home use, we are helping expand Americans’ access to testing, reducing the burden on laboratories and test supplies, and giving Americans more testing options from the comfort and safety of their own homes.”

The announcement today of the first fully at-home OTC COVID-19 diagnostic test follows last month’s [authorization \(/news-events/press-announcements/coronavirus-covid-19-update-fda-authorizes-first-covid-19-test-self-testing-home\)](/news-events/press-announcements/coronavirus-covid-19-update-fda-authorizes-first-covid-19-test-self-testing-home) of the first prescription COVID-19 test for home use and last week’s announcement of the first non-prescription test system, in which a

lab processes the self-collected sample. The FDA has authorized more than 225 diagnostic tests for COVID-19 since the start of the pandemic, including more than 25 tests that allow for home collection of samples, which are then sent to a lab for testing. The Ellume COVID-19 Home Test is the first COVID-19 test that can be used completely at home without a prescription.

“The FDA strongly supports innovation in test development and we have worked tirelessly with test developers to support the shared goal of getting more accurate and reliable tests to Americans who need them. Today is a promising step forward and we are eager to continue advancing additional innovation in COVID-19 testing that the science supports,” said Jeff Shuren, M.D., J.D., director of FDA’s Center for Devices and Radiological Health. “This test, like other antigen tests, is less sensitive and less specific than typical molecular tests run in a lab. However, the fact that it can be used completely at home and return results quickly means that it can play an important role in response to the pandemic.”

Similar to other antigen tests, a small percentage of positive and negative results from this test may be false. Therefore, for patients without symptoms, positive results should be treated as presumptively positive until confirmed by another test as soon as possible. This is especially true if there are fewer infections in a particular community, as false positive results can be more common when antigen tests are used in populations where there is little COVID-19 (low prevalence).

The FDA reminds patients that all tests can experience false negative and false positive results. Individuals with positive results should self-isolate and seek additional care from their health care provider. Individuals who test negative and experience COVID-like symptoms should follow up with their health care provider as negative results do not preclude an individual from SARS-CoV-2 infection.

The Ellume COVID-19 Home Test uses a mid-turbinate nasal swab (sample is collected further back than the usual nasal swab, but not as far back as nasopharyngeal swabs, which are only appropriate for use by a trained health care provider) to detect certain proteins of the virus known as antigens. The Ellume COVID-19 Home Test correctly identified 96% of positive samples and 100% of negative samples in individuals with symptoms. In people without symptoms, the test correctly identified 91% of positive samples and 96% of negative samples. The Ellume COVID-19 Home Test uses an analyzer that connects with a software application on a smartphone to help users perform the test and interpret results. Results are delivered in as little as 20 minutes to individuals via their smartphone. The mobile application

requires individuals to input their zip code and date of birth, with optional fields including name and e-mail address, and reports the results as appropriate to public health authorities to monitor disease prevalence. Ellume expects to produce more than three million tests in January 2021.

The FDA continues to work with test developers to expand access to COVID-19 testing and supports further development of COVID-19 tests that can be used completely at home.

The FDA, an agency within the U.S. Department of Health and Human Services, protects the public health by assuring the safety, effectiveness, and security of human and veterinary drugs, vaccines and other biological products for human use, and medical devices. The agency also is responsible for the safety and security of our nation's food supply, cosmetics, dietary supplements, products that give off electronic radiation, and for regulating tobacco products.

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Inquiries

Media:

✉ [Jim McKinney \(mailto:james.mckinney@fda.hhs.gov\)](mailto:james.mckinney@fda.hhs.gov)

☎ 240-328-7305

Consumer:

☎ 888-INFO-FDA



November 21, 2023

Angela Drysdale
DVP, Regulatory Affairs
Abbott Diagnostics Scarborough, Inc.
10 Southgate Road
Scarborough, ME 04074

Device:	BinaxNOW COVID-19 Antigen Self Test
EUA Number:	EUA210264
Company:	Abbott Diagnostics Scarborough, Inc.
Indication:	Non-prescription home use for the qualitative detection of nucleocapsid protein antigen from SARS-CoV-2 with self-collected anterior nasal (nares) swab samples from individuals aged 15 years or older or adult collected anterior nasal (nares) swab samples from individuals aged two years or older. This test is authorized for individuals with symptoms of COVID-19 within the first seven days of symptom onset when tested at least twice over three days with at least 48 hours between tests, and for individuals without symptoms or other epidemiological reasons to suspect COVID-19, when tested at least three times over five days with at least 48 hours between tests.

Dear Ms. Drysdale:

On March 31, 2021, based on your¹ request, the Food and Drug Administration (FDA) issued an Emergency Use Authorization (EUA) for the BinaxNOW COVID-19 Antigen Self Test pursuant to Section 564 of the Federal Food, Drug, and Cosmetic Act (the Act) (21 U.S.C. §360bbb-3) for the indications stated in the letter.² Based on your requests, FDA granted updates to the authorized labeling on April 1, 2021 and January 7, 2022.^{3,4} Based on your

¹ For ease of reference, this letter will use the term “you” and related terms to refer to Abbott Diagnostics Scarborough, Inc.

² The March 31, 2021, letter authorized your product for the qualitative detection of nucleocapsid protein antigen from SARS-CoV-2 from individuals with or without symptoms or other epidemiological reasons to suspect COVID-19 infection when tested twice over three days with at least 36 hours between tests. This test was authorized for non-prescription home use with self-collected direct anterior nasal (nares) swab samples from individuals aged 15 years or older or adult collected anterior nasal swab samples from individuals aged two years or older.

³ On April 1, 2021, your request was granted via email acknowledgement to include minor updates to the BinaxNOW COVID-19 Antigen Self Test Instructions.

⁴ On January 7, 2022, your request was granted to; (1) update the shelf-life expiration date of the BinaxNOW

requests, the letter was revised and reissued by FDA on August 10, 2021,⁵ November 8, 2021,⁶ April 4, 2022,⁷ and December 23, 2022.⁸ Further, FDA revised the authorized uses and established one additional Condition of Authorization requiring updates to product labeling regarding repeat, or serial, testing, for all currently authorized SARS-CoV-2 antigen tests on November 1, 2022.⁹

On September 20, 2023, you requested that FDA amend this EUA. Based on that request, and having concluded that revising the December 23, 2022, EUA is appropriate to protect the public health or safety under section 564(g)(2)(C) of the Act (21 U.S.C. § 360bbb-3(g)(2)(C)), FDA is

COVID-19 Antigen Self Test to 15 months at room temperature (28–30°C) based on the results of your ongoing stability studies, and (2) add use of an additional nitrocellulose membrane option for manufacturing. FDA also made some minor updates to the Healthcare Provider and Individual Fact Sheets to reflect language use in more recent authorizations.

⁵ On August 10, 2021, the revisions to the March 31, 2020, letter and authorized labeling included: (1) addition of the end user option to use digital electronic-based step-by-step “BinaxNOW COVID-19 Antigen Self Test Instructions” either via a website or by downloading the “NAVICA App” mobile application onto a compatible smartphone, (2) addition of the option to facilitate automatic reporting of test results via the “NAVICA App” mobile application, (3) updates to the performance data to include results of usability studies evaluating the digital electronic-based step-by-step “BinaxNOW COVID-19 Antigen Self Test Instructions,” (4) updates to the Conditions of Authorization to add new Conditions related to circulating variants (Conditions R. and S.) and update timeframe in Condition Q; (5) removal of Conditions of Authorization R. and S. from the March 31, 2021, letter that were fulfilled, and (6) updates to Fact Sheet for Healthcare Providers and Fact Sheet for Individuals to reflect language used in more recent authorizations.

⁶ On November 8, 2021, the revisions to the August 10, 2021, letter and authorized labeling included: (1) updates to the intended use to include use of your product with “*self-collected anterior nasal (nares) swab samples from individuals aged 15 years or older with symptoms of COVID-19 within the first seven days of symptom onset,*” “*adult collected anterior nasal (nares) swab samples from individuals aged two years or older with symptoms of COVID-19 within the first seven days of symptom onset,*” and “*self-collected anterior nasal (nares) swab samples from individuals aged 15 years or older, or adult collected anterior nasal (nares) swab samples from individuals aged two years or older, with or without symptoms or other epidemiological reasons to suspect COVID-19 when tested twice over three days with at least 24 hours (and no more than 48 hours) between tests*”, (2) updates to the outer box labels and Instructions for Use (IFU) limitations section to reflect the updates to the intended use, (3) offering the test kit with 1 test and 10 tests options, in addition to the 2 kit option currently offered, (4) updates to Condition of Authorization Q. to give a 4 month extension, and (5) updates to the letter and fact sheets to reflect the updated intended use and for consistency with language used in more recent authorizations.

⁷ On April 4, 2022, the revisions to the November 8, 2021, letter and authorized labeling included: (1) updates to the outer box labels to include the age range for the test and list a Timer as required but not provided, (2) offering the test kit with 4 tests and 5 tests options, in addition to the 1, 2 and 10 kit options currently offered, (3) updating the date at the top of the Fact Sheet for Healthcare Providers and the Fact Sheet for Individuals to match the date of re-issuance, (4) updates to the letter for consistency with language used in more recent authorizations, and (5) addition of Condition of Authorization M and S (now M and R. below).

⁸ On, December 23, 2022 the revisions to the April 4, 2022, letter and authorized labeling included: (1) incorporating your response to Condition of Authorization (1) of the Repeat Testing Revision Letter dated November 1, 2022 to revise the authorized use(s) as required and described in Appendix A, and making various updates to the authorized labeling as required and described in Appendix B of the letter, (2) discontinuing the Fact Sheet for Individuals as the necessary information is contained in the authorized “BinaxNOW COVID-19 Antigen Self Test Instructions,” (3) deleting Condition of Authorization R. in the April 4, 2022, letter in accordance with the Repeat Testing Revision Letter dated November 1, 2022, and (4) updating the Fact Sheet for Healthcare Providers to reflect the revised intended use, and for consistency with language used in more recent authorizations..

⁹ The Repeat Testing Revision Letter - November 1, 2022, can be accessed at:

<https://www.fda.gov/media/162799/download>.

reissuing the December 23, 2022, letter in its entirety with the revisions incorporated.¹⁰ Accordingly, your product¹¹ is hereby authorized pursuant to section 564 of the Act when used pursuant to the Scope of Authorization (Section II) and Conditions of Authorization (Section IV) of this reissued letter.

On February 4, 2020, as amended on March 15, 2023, pursuant to Section 564(b)(1)(C) of the Act, the Secretary of the Department of Health and Human Services (HHS) determined that there is a public health emergency, or a significant potential for a public health emergency, that affects, or that has a significant potential to affect, national security or the health and security of United States citizens living abroad, and that involves the virus that causes COVID-19. Pursuant to Section 564 of the Act, and on the basis of such determination, the Secretary of HHS then declared that circumstances exist justifying the authorization of emergency use of in vitro diagnostics for detection and/or diagnosis of the virus that causes COVID-19 subject to the terms of any authorization issued under Section 564(a) of the Act.¹²

FDA considered the totality of scientific information available in authorizing the emergency use of your product for the indication above. A summary of the performance information upon which FDA relied is included in the “BinaxNOW COVID-19 Antigen Self Test Healthcare Provider Instructions for Use” (identified below).

Having concluded that the criteria for issuance of this authorization under Section 564(c) of the Act are met, I am authorizing the emergency use of your product, described in the Scope of Authorization of this letter (Section II), subject to the terms of this authorization.

I. Criteria for Issuance of Authorization

I have concluded that the emergency use of your product meets the criteria for issuance of an authorization under Section 564(c) of the Act, because I have concluded that:

1. The SARS-CoV-2 can cause a serious or life-threatening disease or condition, including severe respiratory illness, to humans infected by this virus;
2. Based on the totality of scientific evidence available to FDA, it is reasonable to believe that your product may be effective in diagnosing COVID-19, and that the known and potential benefits of your product when used for diagnosing COVID-19, outweigh the known and potential risks of your product; and

¹⁰ The revisions to the December 23, 2022, letter and authorized labeling include: (1) removing use of and reference to the “NAVICA App” mobile application for use with the BinaxNOW COVID-19 Antigen Self Test, and (2) updating the date of the Fact Sheet for Healthcare Providers to reflect the date of reissuance.

¹¹ For ease of reference, this EUA will use the term “your product” to refer to the BinaxNOW COVID-19 Antigen Self Test used for the indication identified above.

¹² U.S. Department of Health and Human Services, *Determination of a Public Health Emergency and Declaration that Circumstances Exist Justifying Authorizations Pursuant to Section 564(b) of the Federal Food, Drug, and Cosmetic Act*, 21 U.S.C. § 360bbb-3. February 4, 2020. 85 FR 7316 (February 7, 2020). U.S. Department of Health and Human Services, *Amended Determination of a Public Health Emergency or Significant Potential for a Public Health Emergency Pursuant to Section 564(b) of the Federal Food, Drug, and Cosmetic Act*, 21 U.S.C. § 360bbb-3(b). March 15, 2023. 88 FR 16644 (March 20, 2023) (“Amended Determination”).

3. There is no adequate, approved, and available alternative to the emergency use of your product.¹³

II. Scope of Authorization

I have concluded, pursuant to Section 564(d)(1) of the Act, that the scope of this authorization is limited to the indication above.

Authorized Product Details

Your product is a lateral flow immunoassay intended for the qualitative detection of nucleocapsid protein antigen from SARS-CoV-2. This test is authorized for non-prescription home use with self-collected anterior nasal (nares) swab samples from individuals aged 15 years or older or adult collected anterior nasal (nares) swab samples from individuals 2 years or older. This test is authorized for individuals with symptoms of COVID-19 within the first seven days of symptom onset when tested at least twice over three days with at least 48 hours between tests, and for individuals without symptoms or other epidemiological reasons to suspect COVID-19, when tested at least three times over five days with at least 48 hours between tests. Your product does not differentiate between SARS-CoV and SARS-CoV-2.

Results are for the identification of SARS-CoV-2 nucleocapsid protein antigen, which is generally detectable in anterior nasal (nares) swabs during the acute phase of infection. Positive results indicate the presence of viral antigens, but clinical correlation with patient history and other diagnostic information is necessary to determine infection status. Positive results do not rule out bacterial infection or co-infection with other viruses. The agent detected may not be the definite cause of disease. Individuals who test positive with your product should self-isolate and seek follow-up care with their physician or healthcare provider as additional testing may be necessary.

All negative results should be treated as presumptive and confirmation with a molecular assay, if necessary, for patient management, may be performed. Negative results do not rule out SARS-CoV-2 infection and should not be used as the sole basis for treatment or patient management decisions, including infection control measures such as isolating from others and wearing masks. Negative results should be considered in the context of an individual's recent exposures, history and the presence of clinical signs and symptoms consistent with COVID-19. Individuals who test negative and continue to experience COVID-like symptoms of fever, cough and/or shortness of breath may still have SARS-CoV-2 infection and should seek follow up care from their healthcare provider.

Individuals should report their test result obtained with this product to their healthcare provider in order to receive appropriate medical care. All healthcare providers will report all test results they receive from individuals who use the authorized product to relevant public health authorities in accordance with local, state, and federal requirements using appropriate LOINC and SNOMED codes, as defined by the Laboratory In Vitro Diagnostics (LIVD) Test Code Mapping

¹³ No other criteria of issuance have been prescribed by regulation under Section 564(c)(4) of the Act.

for SARS-CoV-2 Tests provided by the Centers for Disease Control and Prevention.

Your product is performed using anterior nasal (nares) swab samples from individuals aged 2 years or older. When using your product, the individual performing the test has the option to follow the “BinaxNOW COVID-19 Antigen Self Test Instructions” provided as a hard copy with the kit or to follow digital electronic-based step-by-step “BinaxNOW COVID-19 Antigen Self Test Instructions” via a website.

The BinaxNOW COVID-19 Antigen Self Test includes the materials, or other authorized materials (as may be requested under Condition L. and M. below), required to collect the anterior nasal (nares) swab sample and perform the test procedure, as described in the “BinaxNOW COVID-19 Antigen Self Test Healthcare Provider Instructions for Use” and the “BinaxNOW COVID-19 Antigen Self Test Instructions.”

Your product includes an internal control test line that must generate the expected result for a test to be considered valid, as outlined in the “BinaxNOW COVID-19 Antigen Self Test Healthcare Provider Instructions for Use” and the “BinaxNOW COVID-19 Antigen Self Test Instructions.”

The labeling entitled “BinaxNOW COVID-19 Antigen Self Test Instructions,” the “BinaxNOW COVID-19 Antigen Self Test Healthcare Provider Instructions for Use,” the website accessible digital electronic “BinaxNOW COVID-19 Antigen Self Test Instructions,” the “BinaxNOW COVID-19 Antigen Self Test” kit box labels¹⁴ (available at <https://www.fda.gov/medical-devices/coronavirus-disease-2019-covid-19-emergency-use-authorizations-medical-devices/in-vitro-diagnostics-euas>), and the following fact sheets pertaining to the emergency use, is required to be made available as set forth in the Conditions of Authorization (Section IV), and are collectively referred to as “authorized labeling:”

- Fact Sheet for Healthcare Professionals¹⁵: Abbott Diagnostics Scarborough, Inc. - The BinaxNOW COVID-19 Antigen Self Test

The above described product, when accompanied by the authorized labeling as set forth in the Conditions of Authorization (Section IV) is authorized to be distributed and used under this EUA, despite the fact that it does not meet certain requirements otherwise required by applicable federal law.

I have concluded, pursuant to Section 564(d)(2) of the Act, that it is reasonable to believe that the known and potential benefits of your product, when used consistent with the Scope of Authorization of this letter (Section II), outweigh the known and potential risks of your product.

¹⁴ “BinaxNOW COVID-19 Antigen Self Test” box labels include boxes for 1, 2, 4, 5 and 10 test kits and “BinaxNOW COVID-19 Antigen Self Test” box labels for additional test kits numbers/options as may be requested, and for which you receive appropriate authorization, in accordance with Condition M. below. BinaxNOW COVID-19 Antigen Self Test numbers/options are described in the “BinaxNOW COVID-19 Antigen Self Test Healthcare Provider Instructions for Use.”

¹⁵ Note that the information typically found in a Fact Sheet for Individuals is contained in the authorized “BinaxNOW COVID-19 Antigen Self Test Instructions” that will be available to end users as set forth in the Conditions of Authorization (Section IV).

I have concluded, pursuant to Section 564(d)(3) of the Act, based on the totality of scientific evidence available to FDA, that it is reasonable to believe that your product may be effective in diagnosing COVID-19, when used consistent with the Scope of Authorization of this letter (Section II), pursuant to Section 564(c)(2)(A) of the Act.

FDA has reviewed the scientific information available to FDA, including the information supporting the conclusions described in Section I above, and concludes that your product (as described in the Scope of Authorization of this letter (Section II)) meets the criteria set forth in Section 564(c) of the Act concerning safety and potential effectiveness.

The emergency use of your product under this EUA must be consistent with, and may not exceed, the terms of this letter, including the Scope of Authorization (Section II) and the Conditions of Authorization (Section IV). Subject to the terms of this EUA and under the circumstances set forth in the Secretary of HHS's determination under Section 564(b)(1)(C) of the Act described above and the Secretary of HHS's corresponding declaration under Section 564(b)(1) of the Act, your product is authorized for the indication above.

III. Waiver of Certain Requirements

I am waiving the following requirements for your product during the duration of this EUA:

- Current good manufacturing practice requirements, including the quality system requirements under 21 CFR Part 820 with respect to the design, manufacture, packaging, labeling, storage, and distribution of your product, but excluding Subpart H (Acceptance Activities, 21 CFR 820.80 and 21 CFR 820.86), Subpart I (Nonconforming Product, 21 CFR 820.90), and Subpart O (Statistical Techniques, 21 CFR 820.250).

IV. Conditions of Authorization

Pursuant to Section 564(e) of the Act, I am establishing the following conditions on this authorization:

Abbott Diagnostics Scarborough, Inc. (You) and Authorized Distributor(s)¹⁶

- A. Your product must comply with the following labeling requirements pursuant to FDA regulations: the intended use statement in 21 CFR 809.10(a)(2), (b)(2); adequate directions for use in 21 U.S.C. 352(f) and 21 CFR 809.10(b)(5), (7), and (8); appropriate limitations on the use of the device including information required under 21 CFR 809.10(a)(4); and any available information regarding performance of the device, including requirements under 21 CFR 809.10(b)(12).
- B. You and authorized distributor(s) must make available the “BinaxNOW COVID-19 Antigen Self Test Instructions” for your product in the shipped kit using the

¹⁶ “Authorized Distributor(s)” are identified by you, Abbott Diagnostics Scarborough, Inc., in your EUA submission as an entity allowed to distribute your product.

“BinaxNOW COVID-19 Antigen Self Test” kit box label (see Footnote 14) and make these instructions electronically available on your website(s).

- C. You and authorized distributor(s) must maintain records of customer complaint files and report to FDA any significant complaints about usability or deviations from the established performance characteristics of which you and authorized distributor(s) become aware.
- D. You and authorized distributor(s) must inform relevant public health authorities of this EUA, including the terms and conditions herein, and any updates made to your product and/or the authorized labeling.
- E. Through a process of inventory control, you and authorized distributor(s) must maintain records of the locations (e.g., pharmacies, doctor’s offices, etc.) to which your product is distributed and the number of tests distributed to each location.
- F. You and authorized distributor(s) must collect information on the performance of your product and have a process in place to track adverse events, including any occurrence of false positive or false negative results and significant deviations from the established performance characteristics of the product of which you become aware and report any such events to FDA in accordance with 21 CFR Part 803. Serious adverse events, especially unexpected biosafety concerns, should immediately be reported to the Division of Microbiology (DMD)/Office of Health Technology 7 (OHT7)-Office of In Vitro Diagnostics/Office of Product Evaluation and Quality (OPEQ)/Center for Devices and Radiological Health (CDRH) (via email: CDRH-EUAREporting@fda.hhs.gov).
- G. You and authorized distributor(s) are authorized to make available additional information relating to the emergency use of your product that is consistent with, and does not exceed, the terms of this letter of authorization.
- H. You and authorized distributor(s) using your product must ensure that any records associated with this EUA are maintained until otherwise notified by FDA. Such records will be made available to FDA for inspection upon request.

Abbott Diagnostics Scarborough, Inc. (You)

- I. You must notify FDA of any authorized distributor(s) of your product, including the name, address, and phone number of any authorized distributor(s).
- J. You must provide authorized distributor(s) with a copy of this EUA and communicate to authorized distributor(s) any subsequent revisions that might be made to this EUA and its authorized accompanying materials, including the authorized labeling.
- K. You must make the authorized “BinaxNOW COVID-19 Antigen Self Test Healthcare Provider Instructions for Use” and the “Fact Sheet for Healthcare Professionals” electronically available on your website. Additionally, you must provide the opportunity

to request a copy of the “BinaxNOW COVID-19 Antigen Self Test Healthcare Provider Instructions for Use” and “Fact Sheet for Healthcare Professionals” in paper form, and after such request, promptly provide the requested labeling at no additional cost.

- L. You may request changes to this EUA for your product, including to the Scope of Authorization (Section II in this letter) or to the authorized labeling, including requests to make available additional authorized labeling specific to an authorized distributor. Such additional labeling may use another name for the product but otherwise must be consistent with the authorized labeling, and shall not exceed the terms of authorization of this letter. Any request for changes to this EUA should be submitted to the DMD/OHT7/OPEQ/CDRH and any changes require appropriate authorization from FDA prior to implementation.
- M. You may request new box labels to allow additional test kits numbers/options for your product. Such additional labeling requests to this EUA should be submitted to and require concurrence of DMD/OHT7/OPEQ/CDRH prior to implementation.
- N. You must comply with the following requirements pursuant to FDA regulations: 21 CFR 820 Subpart H (Acceptance Activities, 21 CFR 820.80 and 21 CFR 820.86), Subpart I (Nonconforming Product, 21 CFR 820.90), and Subpart O (Statistical Techniques, 21 CFR 820.250).
- O. You must have lot release procedures and the lot release procedures, including the study design and statistical power, must ensure that the product released for distribution meet the clinical and analytical performance claimed in the authorized labeling.
- P. If requested by FDA, you must submit your lot release procedures to FDA, including sampling protocols, testing protocols, and acceptance criteria, that you use to release lots of your product for distribution in the U.S. If such lot release procedures are requested by FDA, you must provide them within 48 hours of the request.
- Q. You must evaluate the analytical limit of detection and assess traceability¹⁷ of your product with any FDA-recommended reference material(s), if requested by FDA. After submission to and concurrence with the data by FDA, you will update your labeling to reflect the additional testing, if requested by FDA. Such labeling updates will be made in consultation with, and require concurrence of, DMD/OHT7/OPEQ/CDRH.
- R. You must submit your product for any FDA-recommended independent evaluation to confirm the performance characteristics of your test, if requested by FDA. After submission to and concurrence with the data by FDA, you will update your labeling to reflect the additional testing. Such labeling updates will be made in consultation with, and require concurrence of, DMD/OHT7/OPEQ/CDRH.
- S. You must evaluate the impact of SARS-CoV-2 viral mutations on your product’s performance. Such evaluations must occur on an ongoing basis and must include any

¹⁷ Traceability refers to tracing analytical sensitivity/reactivity back to an FDA-recommended reference material.

additional data analysis that is requested by FDA in response to any performance concerns you or FDA identify during routine evaluation. Additionally, if requested by FDA, you must submit records of these evaluations for FDA review within 48 hours of the request. If your evaluation identifies viral mutations that affect the stated expected performance of your device, you must notify FDA immediately (via email: CDRH-EUA-Reporting@fda.hhs.gov).

- T. If requested by FDA, you must update your labeling within 7 calendar days to include any additional labeling risk mitigations identified by FDA, such as those related to the impact of viral mutations on test performance. Such updates will be made in consultation with, and require concurrence of, DMD/OHT7/OPEQ/CDRH.

Conditions Related to Printed Materials, Advertising and Promotion

- U. All descriptive printed matter, advertising and promotional materials relating to the use of your product shall be consistent with the authorized labeling, as well as the terms set forth in this EUA and meet the requirements set forth in section 502(a), (q)(1), and (r) of the Act, as applicable, and FDA implementing regulations.
- V. No descriptive printed matter, advertising or promotional materials relating to the use of your product may represent or suggest that this test is safe or effective for the detection of SARS-CoV-2.
- W. All descriptive printed matter, advertising and promotional materials relating to the use of your product shall clearly and conspicuously state that:
- This product has not been FDA cleared or approved; but has been authorized by FDA under an EUA;
 - This product has been authorized only for the detection of proteins from SARS- CoV-2, not for any other viruses or pathogens; and,
 - The emergency use of this product is only authorized for the duration of the declaration that circumstances exist justifying the authorization of emergency use of in vitro diagnostics for detection and/or diagnosis of COVID-19 under Section 564(b)(1) of the Federal Food, Drug and Cosmetic Act, 21 U.S.C. § 360bbb-3(b)(1), unless the declaration is terminated or authorization is revoked sooner.

The emergency use of your product as described in this letter of authorization must comply with the conditions and all other terms of this authorization.

V. Duration of Authorization

This EUA will be effective until the declaration that circumstances exist justifying the authorization of the emergency use of in vitro diagnostics for detection and/or diagnosis of

Page 10 – Angela Drysdale, Abbott Diagnostics Scarborough, Inc.

COVID-19 is terminated under Section 564(b)(2) of the Act or the EUA is revoked under Section 564(g) of the Act.

Sincerely,

Jeffrey E. Shuren, M.D., J.D.
Director
Center for Devices and Radiological Health
Food and Drug Administration

Enclosure



November 1, 2022

To: Developers of Antigen In Vitro Diagnostics (IVDs) Authorized for
Emergency Use for Coronavirus Disease 2019 (COVID-19) as of Today's
Date

Re: Revisions Related to Serial (Repeat) Testing for the EUAs of Antigen IVDs

On February 4, 2020, pursuant to Section 564(b)(1)(C) of the Federal Food, Drug, and Cosmetic Act (the Act) (21 U.S.C. § 360bbb-3(b)(1)(C)), the Secretary of the Department of Health and Human Services (HHS) determined that there is a public health emergency that has a significant potential to affect national security or the health and security of United States citizens living abroad, and that involves the virus that causes COVID-19. Pursuant to Section 564 of the Act, and on the basis of such determination, the Secretary of HHS then declared that circumstances exist justifying the authorization of emergency use of in vitro diagnostics for detection and/or diagnosis of the virus that causes COVID-19 subject to the terms of any authorization issued under Section 564(a) of the Act.¹ FDA subsequently authorized the emergency use of numerous in vitro diagnostics (IVDs) for detection and/or diagnosis of SARS-CoV-2, the virus that causes COVID-19.²

Pursuant to Section 564 of the Act, and in response to new data regarding performance of antigen tests from a study assessing at-home COVID-19 antigen test performance ("the antigen

¹ U.S. Department of Health and Human Services, *Determination of a Public Health Emergency and Declaration that Circumstances Exist Justifying Authorizations Pursuant to Section 564(b) of the Federal Food, Drug, and Cosmetic Act*, 21 U.S.C. § 360bbb-3. 85 FR 7316 (February 7, 2020)

² In Vitro Diagnostics EUAs: <https://www.fda.gov/medical-devices/coronavirus-disease-2019-covid-19-emergency-use-authorizations-medical-devices/in-vitro-diagnostics-euas>, includes links to tables of currently authorized IVD EUAs for SARS-CoV-2.

study”),³ FDA is revising the authorized uses⁴ and requiring updates to product labeling of all tests that are within the scope of this letter. This revision also establishes one additional Condition of Authorization, and eliminates one Condition of Authorization, on EUAs that are within the scope of this revision (Section I).

The additional condition of authorization established by this revision concerns updates to the authorized labeling⁵ to reflect the revised authorized use for tests that are within the scope of this revision. FDA’s determination that the Condition of Authorization established by this revision is necessary or appropriate to protect the public health is based on the available scientific evidence and FDA’s continuing efforts to evaluate the performance of authorized antigen IVDs with respect to the use of serial testing. The eliminated condition of authorization concerns the collection of additional data to evaluate the performance of authorized antigen IVDs with respect to the use of serial testing. FDA’s determination that the Condition of Authorization eliminated by this revision is no longer necessary or appropriate to protect the public health in light of the antigen study.

Having concluded that the revisions to the EUAs of tests that are within the scope of this letter (section I) are appropriate to protect the public health or safety, I am hereby revising all such EUAs pursuant to Section 564(g)(2)(C), including to revise the authorized use and to establish the additional condition set forth in this letter as permitted by Section 564(e) of the Act. This

³ COVID-19 antigen tests are less likely to detect the SARS-CoV-2 virus than molecular tests, such as polymerase chain reaction (PCR) tests. However, repeat, or serial, antigen testing has been shown to improve the ability of an antigen test to detect the virus in time for an individual to take actions to contain the spread of disease. To investigate the performance of SARS-CoV-2 antigen serial testing and generate data to support regulatory decisions, FDA collaborated with the National Institutes for Health (NIH) and the University of Massachusetts Chan Medical School to design and implement a study (*Finding a Needle in the Haystack: Design and Implementation of a Digital Site-less Clinical Study of Serial Rapid Antigen Testing to Identify Asymptomatic SARS-CoV-2 Infection* - <https://www.medrxiv.org/content/10.1101/2022.08.04.22278274v1>, referred to as “the antigen study” in this letter) to assess at-home COVID-19 antigen test performance. Results from the antigen study (*Performance of Screening for SARS-CoV-2 using Rapid Antigen Tests to Detect Incidence of Symptomatic and Asymptomatic SARS-CoV-2 Infection: findings from the Test Us at Home prospective cohort study* - <https://www.medrxiv.org/content/10.1101/2022.08.05.22278466v1>) show that repeat testing over multiple days improves test performance and increases the likelihood that an at-home COVID-19 antigen test will accurately detect an infection. These results have informed the FDA's general understanding that repeat testing after a negative result with an at-home COVID-19 antigen test reduces the risk of a false negative result. On August 11, 2022, based on the results of the antigen study, FDA advised individuals to perform repeat, or serial, testing following a negative result on any at-home COVID-19 antigen test, to reduce the risk an infection may be missed (false negative result) and to help prevent individuals from unknowingly spreading the SARS-CoV-2 virus to others. The FDA recommended repeat testing following a negative result whether or not an individual has COVID-19 symptoms. (<https://www.fda.gov/medical-devices/safety-communications/home-covid-19-antigen-tests-take-steps-reduce-your-risk-false-negative-fda-safety-communication>).

⁴ Generally, the authorized uses of antigen tests are described in, among other locations, the “indication” discussion and “authorized product detail” section in the letters of authorization, and in an “intended use” section in the authorized labeling.

⁵ Authorized labeling impacted by the additional condition of authorization established by this letter includes some combination of the following documents: Instructions for Use (IFU), Quick Reference Instructions (QRI), laboratory Standard Operating Procedures (SOPs), electronic labeling and applications, and Fact Sheets. Note that the Fact Sheets are generated by FDA who will update the documents after it receives the supplement request to update the test’s other labeling consistent with this revision.

action is based on the available scientific evidence⁶ on the impact of serial testing on the performance of SARS-CoV-2 antigen tests.

I. Scope of this Revision

This letter revises all current EUAs for antigen SARS-CoV-2 IVD devices⁷ as of today's date by:

- (1) revising the authorized use to be for serial testing at least twice over three days for individuals with symptoms of COVID-19 and, for tests previously authorized for testing individuals without symptoms, revising the authorized use to be for serial testing at least three times over five days for individuals without symptoms of COVID-19, as set forth in Appendix A of this letter,
- (2) establishing a new condition of authorization, as set forth in Section III of this letter, on such authorizations, and
- (3) eliminating a condition of authorization, as set forth in Section III of this letter, on such authorizations.

This revision does not apply to EUAs for non-antigen based authorized assays (e.g., molecular, serology), EUAs for authorized IL-6 assays, EUAs for standalone specimen collection devices, or EUAs for standalone home collection kits.

All updated labeling will be added to FDA's webpage and posted with the EUA after it is submitted to FDA as required by Condition of Authorization (1) of this letter.

II. Waiver of Certain Requirements

This revision does not change the waiver of any requirements included in the EUAs being revised.

III. Revisions to Conditions of Authorization

- A. Pursuant to Section 564(e) of the Act, I am establishing the additional condition below with respect to repeat testing on all authorized tests within this letter's scope.

Developer (You)

- (1) You must update your authorized labeling to reflect the revised authorized use in Appendix A and as set forth in Appendix B of this letter by submitting your proposed updated labeling to FDA as a supplement to your EUA within 10 business days of today's date, unless otherwise agreed to by the Division of Microbiology (DMD)/Office of Health Technology 7 (OHT7)-Office of In Vitro Diagnostics

⁶ *Performance of Screening for SARS-CoV-2 using Rapid Antigen Tests to Detect Incidence of Symptomatic and Asymptomatic SARS-CoV-2 Infection: findings from the Test Us at Home prospective cohort study* - <https://www.medrxiv.org/content/10.1101/2022.08.05.22278466v1>

⁷ Please see the following link for currently authorized Antigen IVD devices for SARS-CoV-2: <https://www.fda.gov/medical-devices/coronavirus-disease-2019-covid-19-emergency-use-authorizations-medical-devices/in-vitro-diagnostics-euas-antigen-diagnostic-tests-sars-cov-2>.

/Office of Product Evaluation and Quality (OPEQ)/Center for Devices and Radiological Health (CDRH). Following FDA's concurrence with the supplement, you must update your electronic labeling and electronic/mobile applications (apps) within 20 business days. You must update your paper labeling within 30 business days of FDA's concurrence and all tests distributed subsequently must be accompanied by the updated paper labeling.

- B. Pursuant to section 564(g)(2)(C) of the Act, and in consideration of the establishment of Condition of Authorization (1) above, I am eliminating the following condition (or similar condition) from EUAs within the scope (section I above):

Developer (You)

You must evaluate the clinical performance of your product to support the serial screening claim in an FDA agreed upon post authorization clinical evaluation study within 4 months of the date of this letter (unless otherwise agreed to with DMD/OHT7/OPEQ/CDRH). After submission to and concurrence with the data by FDA, you must update the authorized labeling to reflect the additional testing. Such labeling updates will be made in consultation with, and require concurrence of, DMD/OHT7/OPEQ/CDRH.

Sincerely,

Namandjé N. Bumpus, Ph.D.
Chief Scientist
Food and Drug Administration

Enclosure

Appendix A
Revised Authorized Uses

The authorized uses for tests that are within the scope (Section I) of this letter are revised as follows:

- 1) Where a test was previously authorized for testing of symptomatic individuals (e.g., within the first [*number specific to each test*] days of symptom onset), the test is now authorized for use at least twice over three days with at least 48 hours between tests.
- 2) Where a test was previously authorized for testing of asymptomatic individuals (e.g., individuals without symptoms or other epidemiological reasons to suspect COVID-19), the test is now authorized for use at least three times over five days with at least 48 hours between tests.

Appendix B

Required Changes to Authorized Labeling

As required by Condition of Authorization (1), you must update your authorized labeling to include the labeling elements, sections, and statements below:

1. Intended Use:⁸

In addition to name, technology, analyte specific information and other validated claims, the Intended Use must include the updated frequency of testing for symptomatic and asymptomatic individuals (as applicable).

- For tests referenced in Appendix A, #1, this means including language that the test is for serial testing for use at least twice over three days with at least 48 hours between tests;
- For tests referenced in Appendix A, #2, this means including language that the test is for serial testing for use at least three times over five days with at least 48 hours between tests and removing language regarding testing at least twice over two or three days with at least 24 and no more than 36 hours between tests.

In addition, the Intended Use must include the following statements:

- “negative results are presumptive”
- “The [*Test Name*] is only for in vitro diagnostic use under the Food and Drug Administration’s Emergency Use Authorization. This product has not been FDA cleared or approved.”

2. Outer Box and Subassembly Labeling (as applicable):

- Expiration date (sticker): Based on component of kit with the earliest expiration date
- Summary of box contents
- Items necessary to use the test but not provided in the test kit: [*e.g., access to computer/smartphone, internet, email account*]
- Storage temperature
- Summary of how the kit works

Statements that must be present on the outer box:

- For Emergency Use Authorization (EUA) only
- For in vitro diagnostic use
- In the USA, this product has not been FDA cleared or approved, but has been authorized by FDA under an EUA.

⁸ As discussed in footnote 4, generally the authorized uses of antigen tests are described in, among other locations, an “intended use” section in the authorized labeling. If such uses are discussed elsewhere in the authorized labeling, you must also make appropriate updates to that labeling to reflect the revised authorized uses.

- This product has been authorized only for the detection of proteins from SARS-CoV-2 [*for multi-analyte tests please insert the additional on-panel analytes, e.g., influenza A/B*], not for any other viruses or pathogens.
- The emergency use of this product is only authorized for the duration of the declaration that circumstances exist justifying the authorization of emergency use of in vitro diagnostics for detection and/or diagnosis of COVID-19 under Section 564(b)(1) of the Federal Food, Drug, and Cosmetic Act, 21 U.S.C. § 360bbb-3(b)(1), unless the declaration is terminated or authorization is revoked sooner.
- Determining a negative result requires multiple tests. You may need to purchase additional tests to perform serial (repeat) testing. This test is more likely to give you a false negative result when you have COVID-19 than a lab-based molecular test.

3. Instructions for Use

a. Lay User Labeling (applicable only to tests authorized for home use)

i. Test Interpretation

Repeat testing is needed to improve test accuracy. Please follow the table below when interpreting test results for COVID-19.

Status on First Day of Testing	First Result Day 1	Second Result Day 3	Third Result Day 5	Interpretation
With Symptoms	Positive	N/A	N/A	Positive for COVID-19
	Negative	Positive	N/A	Positive for COVID-19
	Negative	Negative	N/A	Negative for COVID-19
Without Symptoms	Positive	N/A	N/A	Positive for COVID-19
	Negative	Positive	N/A	Positive for COVID-19
	Negative	Negative	Positive	Positive for COVID-19
	Negative	Negative	Negative	Negative for COVID-19

Results should be considered in the context of an individual's recent exposures, history, and the presence of clinical signs and symptoms consistent with COVID-19.

COVID-19 Positive (+)

If the Control (C) line and the Test (T) line are visible, the test is positive. Any faint visible [*color*] test (T) line with the control line (C) should be read as positive.

You do not need to perform repeat testing if you have a positive result at any time.

A positive test result means that the virus that causes COVID-19 was detected in your sample and it is very likely you have COVID-19 and are contagious. Please contact your doctor/primary care physician or your local health authority immediately and adhere to the local guidelines regarding self-isolation. There is a very small chance that this test can give a positive result that is incorrect (a false positive).

COVID-19 Negative (-)

If the Control (C) line is visible, but the Test (T) line is not visible, the test is negative.

To increase the chance that the negative result for COVID-19 is accurate, you should:

- **Test again in 48 hours if you have symptoms on the first day of testing.**
- **Test 2 more times at least 48 hours apart if you do not have symptoms on the first day of testing.**

A negative test result indicates that the virus that causes COVID-19 was not detected in your sample. A negative result is presumptive, meaning it is not certain that you do not have COVID-19. You may still have COVID-19 and you may still be contagious. There is a higher chance of false negative results with antigen tests compared to laboratory-based tests such as PCR. If you test negative and continue to experience COVID-19-like symptoms, (e.g., fever, cough, and/or shortness of breath) you should seek follow up care with your health care provider.

Invalid

If the control (C) line is not visible, the test is invalid. Re-test with a new swab and new test device.

ii. How to Use This Test

The Lay User labeling must include a specific section “How to Use this Test” with the following information:

- Serial testing should be performed in all individuals with negative results; individuals with symptoms of COVID-19 and initial negative results should be tested again after 48 hours. Individuals without symptoms of COVID-19, and with initial negative results, should be tested again after 48 hours and, if the 2nd test is also negative, a 3rd time after an additional 48 hours. You may need to purchase additional tests to perform this serial (repeat) testing.
- If you test negative but continue to have symptoms of COVID-19, and both your first and second tests are negative, you may not have COVID-19, however you should follow-up with your healthcare provider.
- If your test is positive, then proteins from the virus that causes COVID-19 have been found in your sample and you likely have COVID-19.

iii. Warnings, Precautions, and Safety Information

The Lay User labeling must include the following warnings, precautions and/or safety information:

- Read all instructions carefully before performing the test. Failure to follow the instructions may result in inaccurate test results.
- In the USA, this product has not been FDA cleared or approved, but has been authorized by FDA under an Emergency Use Authorization. This product has been authorized only for the detection of proteins from SARS-CoV-2, not for any other viruses or pathogens. The emergency use of this product is only authorized for the duration of the declaration that circumstances exist justifying the authorization of emergency use of in vitro diagnostics for detection and/or diagnosis of COVID-19 under Section 564(b)(1) of the Federal Food, Drug, and Cosmetic Act, 21 U.S.C. § 360bbb-3(b)(1), unless the declaration is terminated or authorization is revoked sooner.
- **Serial testing should be performed in individuals with negative results at least twice over three days (with 48 hours between tests) for symptomatic individuals and three times over five days (with at least 48 hours between tests) for asymptomatic individuals. You may need to purchase additional tests to perform this serial (repeat) testing.**
- An anterior nasal swab sample can be self-collected by an individual age [X] years and older. Children age 2 to [X] years should be tested by an adult.
- Do not use on anyone under 2 years of age.
- Wear a safety mask or other face-covering when collecting a specimen from a child or another individual.
- Do not use if any of the test kit contents or packaging is damaged.
- Test components are single-use. Do not re-use.
- Do not use kit past its expiration date.
- Do not touch the swab tip.
- Once opened, the test card should be used within [X] minutes.
- Do not read test results before [X] minutes or after [Y] minutes. Results read before [X] minutes or after [Y] minutes may lead to a false positive, false negative, or invalid result.
- If applicable: Keep testing kit and kit components away from children and pets before and after use. Avoid contact with your [e.g., *skin, eyes, nose, or mouth*]. Do not ingest any kit components. The reagent solution contains harmful chemicals (see table below). If the solution contacts your [e.g., *e.g., skin, eyes, nose, or mouth*], flush with large amounts of water.

If irritation persists, seek medical advice: <https://www.poisonhelp.org> or 1-800-222-1222. *[Note: Please do not use the following website in your labeling: <https://www.poison.org/contact-us>. Also, please populate the following table as appropriate for your device and per FDA toxicological assessment of your device:]*

Chemical Name	GHS Code for each Ingredient	Concentrations
e.g., Microcide III	H315, skin irritation	0.2%

- For more information on EUAs please visit: <https://www.fda.gov/emergency-preparedness-and-response/mcm-legal-regulatory-and-policy-framework/emergency-use-authorization>
- For the most up to date information on COVID-19, please visit: www.cdc.gov/COVID19

iv. Limitations

- There is a higher chance of false negative results with antigen tests than with laboratory-based molecular tests due to the sensitivity of the test technology. This means that there is a higher chance this test will give a false negative result in an individual with COVID-19 as compared to a molecular test, especially in samples with low viral load.
- The performance of this test was established based on the evaluation of a limited number of clinical specimens collected between [*month, year and month, year*]. The clinical performance has not been established for all circulating variants but is anticipated to be reflective of the prevalent variants in circulation at the time and location of the clinical evaluation. Performance at the time of testing may vary depending on the variants circulating, including newly emerging strains of SARS-CoV-2 and their prevalence, which change over time.
- All COVID-19 antigen test negative results are presumptive and confirmation with a molecular assay may be necessary. If you continue to have symptoms of COVID-19, and both your first and second tests are negative, you may not have COVID-19, however you should follow-up with a healthcare provider.
- If the test is positive, then proteins from the virus that causes COVID-19 have been found in the sample and you likely have COVID-19.
- This test is read visually and has not been validated for use by those with impaired vision or color-impaired vision.
- Incorrect test results may occur if a specimen is incorrectly collected or handled.

v. Frequently Asked Questions (FAQ):

The following FAQ section should be added to the Lay User IFU (and, if applicable, the QRI) for those Lay User IFUs that also contain the information typically found in a Patient Fact Sheet for OTC tests:

WHAT ARE THE KNOWN AND POTENTIAL RISKS AND BENEFITS OF THE TEST?

Potential risks include:

- Possible discomfort during sample collection.
- Possible incorrect test result (see Warnings and Result Interpretation sections for more information).

Potential benefits include:

- The results, along with other information, can help you and your healthcare provider make informed recommendations about your care.
- The results of this test may help limit the potential spread of COVID-19 to your family and others in your community.

For more information on EUAs go here: <https://www.fda.gov/emergency-preparedness-and-response/mcm-legal-regulatory-and-policy-framework/emergency-use-authorization>

WHAT IS THE DIFFERENCE BETWEEN AN ANTIGEN AND MOLECULAR TEST?

There are different kinds of tests for the SARS-CoV-2 virus that causes COVID-19. Molecular tests detect genetic material from the virus. Antigen tests, such as the [*Test Name*], detect proteins from the virus. Due to the lower sensitivity of antigen tests, there is a higher chance this test will give you a false negative result when you have COVID-19 than a molecular test would.

HOW ACCURATE IS THIS TEST?

Clinical studies have shown that antigen tests more accurately determine whether you are infected with the virus that causes COVID-19 when taken multiple times across several days. Repeat testing improves test accuracy. This serial testing approach is recommended to minimize the risk of incorrect results. For more information on the performance of the test and how the performance may apply to you, please refer to the performance data in the Healthcare Provider Instructions for Use (IFU), available at [*insert developer's website's address*].

WHAT IF I HAVE A POSITIVE TEST RESULT?

A positive result means that it is very likely you have COVID-19 because proteins from the virus that causes COVID-19 were found in your sample. You should self-isolate from others and contact a healthcare provider for medical advice about your positive result.

WHAT IF I HAVE A NEGATIVE TEST RESULT?

A negative test result indicates that antigens from the virus that causes COVID-19 were not detected in your sample. However, if you have symptoms of COVID-19, and your first test is negative, you should test again in 48 hours since antigen tests are not as sensitive as molecular tests. If you do not have symptoms and received a negative result, you should test at least two more times with 48 hours in between tests for a total of three tests. If you have a negative result, it does not rule out SARS-CoV-2 infection; you may still be infected and you may still infect others. It is important that you work with your healthcare provider to help you understand the next steps you should take.

WHAT DOES AN INVALID TEST RESULT MEAN?

An invalid result means the test was not able to tell if you have COVID-19 or not. If the test is invalid, a new swab should be used to collect a new nasal specimen and you should test again with a new test.

IMPORTANT

Do not use this test as the only guide to manage your illness. Consult your healthcare provider if your symptoms persist or become more severe.

Individuals should provide all results obtained with this product to their healthcare provider.

b. HCP Labeling

i. Test Interpretation

Repeat testing is needed to improve test accuracy. Please follow the table below when interpreting test results.

Status on first day of Testing	First Result Day 1	Second Result Day 3	Third Result Day 5	Interpretation
With Symptoms	Positive	N/A	N/A	Positive for COVID-19
	Negative	Positive	N/A	Positive for COVID-19
	Negative	Negative	N/A	Negative for COVID-19
Without Symptoms	Positive	N/A	N/A	Positive for COVID-19
	Negative	Positive	N/A	Positive for COVID-19
	Negative	Negative	Positive	Positive for COVID-19
	Negative	Negative	Negative	Negative for COVID-19

Results should be considered in the context of an individual's recent exposures, history, and the presence of clinical signs and symptoms consistent with COVID-19.

COVID-19 Positive (+)

If the Control (C) line and the Test (T) line are visible, the test is positive. Any faint visible [color] test (T) line with the control line (C) should be read as positive.

Repeat testing does not need to be performed if patients have a positive result at any time.

A positive test result means that the virus that causes COVID-19 was detected in the sample, and it is very likely the individual has COVID-19 and is contagious. Please contact the patient's doctor/primary care physician (if applicable) and the local health authority immediately and instruct your patient to adhere to the local guidelines regarding self-

isolation. There is a very small chance that this test can give a positive result that is incorrect (a false positive).

Positive results do not rule out bacterial infection or co-infection with other viruses. The agent detected may not be the definite cause of disease. Individuals who test positive with the [Test Name] should self-isolate and seek follow up care with their physician or healthcare provider as additional confirmatory testing with a molecular test for positive results may also be necessary, if there is a low likelihood of COVID-19, such as in individuals without known exposures to COVID-19 or residing in communities with low prevalence of infection.

COVID-19 Negative (-)

If the Control (C) line is visible, but the Test (T) line is not visible, the test is negative.

To increase the chance that the negative result for COVID-19 is accurate, you should:

- **Test again in 48 hours if the individual has symptoms on the first day of testing.**
- **Test 2 more times at least 48 hours apart if the individual does not have symptoms on the first day of testing.**

A negative test result indicates that the virus that causes COVID-19 was not detected in the sample. A negative result does not rule out COVID-19. There is a higher chance of false negative results with antigen tests compared to laboratory-based tests such as PCR tests. If the test is negative but COVID-19-like symptoms, e.g., fever, cough, and/or shortness of breath continue, follow up testing for SARS-CoV-2 with a molecular test or testing for other respiratory disease should be considered. If applicable, seek follow up care with the primary health care provider.

All negative results should be treated as presumptive and confirmation with a molecular assay may be necessary if there is a high likelihood of SARS-CoV-2 infection, such as in an individual with a close contact with COVID-19 or with suspected exposure to COVID-19 or in communities with high prevalence of infection. Negative results do not rule out SARS-CoV-2 infection and should not be used as the sole basis for treatment or patient management decisions, including infection control decisions.

Invalid

If the control (C) line is not visible, the test is invalid. Re-test with a new swab and new test device.

ii. Warnings, Precautions, and Safety Information

The following Warnings, Precautions, and Safety Information should be included in the full HCP IFU; in addition, the warning statements of the first three bullet points below should also be included in the Quick Reference Guide (QRI) of all tests authorized for point-of-care (PoC) use.

- Read all instructions carefully before performing the test. Failure to follow the instructions may result in inaccurate test results.
- In the USA, this product has not been FDA cleared or approved, but has been authorized by FDA under an Emergency Use Authorization. This product has been authorized only for the detection of proteins from SARS-CoV-2 [*for multi-analyte tests please insert the additional on-panel analytes, e.g., influenza A/B*], not for any other viruses or pathogens. The emergency use of this product is only authorized for the duration of the declaration that circumstances exist justifying the authorization of emergency use of in vitro diagnostics for detection and/or diagnosis of COVID-19 under Section 564(b)(1) of the Federal Food, Drug, and Cosmetic Act, 21 U.S.C. § 360bbb-3(b)(1), unless the declaration is terminated or authorization is revoked sooner.
- **Serial testing should be performed in individuals with negative results at least twice over three days (with 48 hours between tests) for symptomatic individuals and three times over five days (with at least 48 hours between tests) for asymptomatic individuals. You may need to purchase additional tests to perform this serial (repeat) testing.**
- An anterior nasal swab sample can be self-collected by an individual age [X] years and older. Children age 2 to [X] years should be tested by an adult.
- Do not use on anyone under 2 years of age.
- Wear a safety mask or other face-covering when collecting a specimen from a child or another individual.
- Do not use if any of the test kit contents or packaging is damaged.
- Test components are single-use. Do not re-use.
- Do not use kit past its expiration date.
- Do not touch the swab tip.
- Once opened, the test card should be used within [X] minutes.
- **Do not read test results before [X] minutes or after [Y] minutes. Results read before [X] minutes or after [Y] minutes may lead to a false positive, false negative, or invalid result.**
- **Keep testing kit and kit components away from children and pets before and after use. Avoid contact with your [e.g., skin, eyes, nose, or mouth]. Do not ingest any kit components. The reagent solution contains harmful chemicals (see table below). If the solution contacts your [e.g., skin, eyes, nose, or mouth], flush with large amounts of water. If irritation persists, seek medical advice: <https://www.poisonhelp.org> or 1-800-222-1222.**

[Note: Please do not use the following website in your labeling:

<https://www.poison.org/contact-us>. Also, please populate the following table as appropriate for your device and per FDA toxicological assessment of your device:]

Chemical Name	GHS Code for each Ingredient	Concentrations
e.g., Microcide III	H315, skin irritation	0.2%

- For more information on EUAs please visit: <https://www.fda.gov/emergency-preparedness-and-response/mcm-legal-regulatory-and-policy-framework/emergency-use-authorization>
- For the most up to date information on COVID-19, please visit: www.cdc.gov/COVID19

iii. Limitations

- The performance of this test was established based on the evaluation of a limited number of clinical specimens collected between [*month, year and month, year*]. The clinical performance has not been established for all circulating variants but is anticipated to be reflective of the prevalent variants in circulation at the time and location of the clinical evaluation. Performance at the time of testing may vary depending on the variants circulating, including newly emerging strains of SARS-CoV-2 and their prevalence, which change over time.
- There is a higher chance of false negative results with antigen tests than with laboratory-based molecular tests due to the sensitivity of the test technology. This means that there is a higher chance this test will give a false negative result in an individual with COVID-19 as compared to a molecular test, especially in samples with low viral load.
- All COVID-19 antigen test negative results are presumptive and confirmation with a molecular assay may be necessary.
- If the patient continues to have symptoms of COVID-19, and both the patient's first and second tests are negative, the patient may not have COVID-19, however additional follow-up may be needed.
- If the test is positive, then proteins from the virus that causes COVID-19 have been found in the sample and the individual likely has COVID-19.
- This test is read visually and has not been validated for use by those with impaired vision or color-impaired vision.
- Incorrect test results may occur if a specimen is incorrectly collected or handled.
- This test detects both viable (live) and nonviable SARS-CoV-2. Test performance depends on the amount of virus (antigens) in the sample and may or may not correlate with viral culture results performed on the same sample.

iv. Performance Section of HCP labeling

A prospective clinical study was conducted between January 2021 and May 2022 as a component of the Rapid Acceleration of Diagnostics (RADx) initiative from the National Institutes of Health (NIH). A total of 7,361 individuals were enrolled via a decentralized clinical study design, with a broad geographical representation of the United States. Per inclusion criteria, all individuals were asymptomatic upon enrollment in the study and at least 14 days prior to it and did not have a SARS-CoV-2 infection in the three months prior to enrollment. Participants were assigned to one of three EUA authorized SARS-CoV-2 OTC rapid antigen tests to conduct serial testing (every 48 hours) for 15 days. If an antigen test was positive, the serial-antigen testing result is considered positive.

At each rapid antigen testing time point, study subjects also collected a nasal swab for comparator testing using a home collection kit (using a 15-minute normalization window between swabs). SARS-CoV-2 infection status was determined by a composite comparator method on the day of the first antigen test, using at least two highly sensitive EUA RT-PCRs. If results of the first two molecular test were discordant a third highly sensitive EUA RT-PCR test was performed, and the final test result was based upon the majority rule.

Study participants reported symptom status throughout the study using the MyDataHelps app. Two-day serial antigen testing is defined as performing two antigen tests 36 - 48 hours apart. Three-day serial antigen testing is defined as performing three antigen tests over five days with at least 48 hours between each test.

Out of the 7,361 participants enrolled in the study, 5,609 were eligible for analysis. Among eligible participants, 154 tested positive for SARS-CoV-2 infection based on RT-PCR, of which 97 (62%) were asymptomatic on the first day of their infection, whereas 57 (39%) reported symptoms on the first day of infection. Pre-symptomatic subjects were included in the positive percent agreement (PPA) of asymptomatic individuals, if they were asymptomatic on the first day of antigen testing, regardless of whether they developed symptoms at any time after the first day of testing.

Performance of the antigen test with serial testing in individuals is described in Table YYY.

Table YYY: Data establishing PPA of COVID-19 antigen serial testing compared to the molecular comparator single day testing throughout the course of infection with serial testing. Data is from all antigen tests in study combined.

DAYS AFTER FIRST PCR POSITIVE TEST RESULT	ASYMPTOMATIC ON FIRST DAY OF TESTING			SYMPTOMATIC ON FIRST DAY OF TESTING		
	Ag Positive / PCR Positive (Antigen Test Performance % PPA)					
	1 Test	2 Tests	3 Tests	1 Test	2 Tests	3 Tests
0	9/97	35/89	44/78	34/57	47/51	44/47
	(9.3%)	(39.3%)	(56.4%)	(59.6%)	(92.2%)	(93.6%)
2	17/34	23/34	25/32	58/62	59/60	43/43
	(50.0%)	(67.6%)	(78.1%)	(93.5%)	(98.3%)	(100%)
4	16/21	15/20	13/15	55/58	53/54	39/40
	(76.2%)	(75.0%)	(86.7%)	(94.8%)	(98.1%)	(97.5%)
6	20/28	21/27	16/18	27/34	26/33	22/27
	(71.4%)	(77.8%)	(88.9%)	(79.4%)	(78.8%)	(81.5%)
8	13/23	13/22	4/11	12/17	12/17	7/11
	(56.5%)	(59.1%)	(36.4%)	(70.6%)	(70.6%)	(63.6%)
10	5/9	5/8		4/9	3/7	
	(55.6%)	(62.5%)		(44.4%)	(42.9%)	

1 Test = one (1) test performed on the noted days after first PCR positive test result. Day 0 is the first day of documented infection with SARS-CoV-2.

2 Tests = two (2) tests performed an average of 48 hours apart. The first test performed on the indicated day and the second test performed 48 hours later.

3 Tests = three (3) tests performance an average of 48 hours apart. The first test performed on the indicated day, the second test performed 48 hours later, and a final test performed 48 hours after the second test.

4. Fact Sheets

The Fact Sheet for Healthcare Providers (HCPs) and Fact Sheet for Patients and Fact Sheet for Individuals will be updated by FDA consistent with this revision, and FDA will provide them to you.

5. Electronic Applications (e.g., Cell Phone Apps)

Electronic applications (apps) should include a prominently placed warning for the requirement of repeat testing after a negative test result. App related software should be updated to accommodate the following minimal information:

- Test interpretation as outlined in section 3. a. i. above.
- How to Use This Test as outlined in section 3. a. ii. above
- Warning Statements:
 - In the USA, this product has not been FDA cleared or approved, but has been authorized by FDA under an Emergency Use Authorization. This product has been authorized only for the detection of proteins from SARS-CoV-2, not for any other viruses or pathogens. The emergency use of this product is only authorized for the duration of the declaration that circumstances exist justifying the authorization of emergency use of in vitro diagnostics for detection and/or diagnosis of COVID-19 under Section 564(b)(1) of the Federal Food, Drug, and Cosmetic Act, 21 U.S.C. § 360bbb-3(b)(1), unless the declaration is terminated or authorization is revoked sooner.
 - **Serial testing should be performed in individuals with negative results at least twice over three days (with 48 hours between tests) for symptomatic individuals and three times over five days (with at least 48 hours between tests) for asymptomatic individuals. You may need to purchase additional tests to perform this serial (repeat) testing.**
 - There is a higher chance of false negative results with antigen tests than with laboratory-based molecular tests due to the sensitivity of the test technology. This means that there is a higher chance this test will give a false negative result in an individual with COVID-19 as compared to a molecular test, especially in samples with low viral load.
 - All COVID-19 antigen test negative results are presumptive and confirmation with a molecular assay may be necessary. If you continue to have symptoms of

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COVID-19, and both your first and second tests are negative, you may not have COVID-19, however you should follow-up with a healthcare provider.

- This test is read visually and has not been validated for use by those with impaired vision or color-impaired vision.
- A link to an FAQ document as described in section 3. a. v. above.



June 6, 2023

Cue Health Inc.
Roderick Castillo
Sr. Director Regulatory and Clinical Affairs
4980 Carroll Canyon Road Suite 100
San Diego, CA 92121 USA

Re: DEN220028

Trade/Device Name: Cue COVID-19 Molecular Test
Regulation Number: 21 CFR 866.3984
Regulation Name Over-the-counter test to detect SARS-CoV-2 from clinical specimens
Regulatory Class: Class II
Product Code: QWB
Dated: April 27, 2022
Received: May 2, 2022

Dear Roderick Castillo:

The Center for Devices and Radiological Health (CDRH) of the Food and Drug Administration (FDA) has completed its review of your De Novo request for classification of the Cue COVID-19 Molecular Test, an over-the-counter device with the following indications for use:

The Cue COVID-19 Molecular Test is a nucleic acid amplification assay that is used with the Cue Monitoring System (Cue Cartridge Reader) for the rapid, qualitative detection of SARS-CoV-2 nucleic acid directly in anterior nasal swab specimens from individuals with signs and symptoms of COVID-19 (i.e., symptomatic).

A negative test result is presumptive, and it is recommended these results be confirmed by a lab-based molecular SARS-CoV-2 assay if necessary for patient management. Negative results do not preclude SARS-CoV-2 infections and should not be used as the sole basis for treatment.

Positive results do not rule out co-infection with other respiratory pathogens.

This test is not a substitute for visits to a healthcare provider or appropriate follow-up and should not be used to determine any treatments without provider supervision.

This test is intended to be sold over-the-counter (OTC) for testing of individuals 18 years of age and older.

FDA concludes that this device should be classified into Class II. This order, therefore, classifies the Cue COVID-19 Molecular Test, and substantially equivalent devices of this generic type, into Class II under the generic name over-the-counter test to detect SARS-CoV-2 from clinical specimens.



FDA identifies this generic type of device as:

Over-the-counter test to detect SARS-CoV-2 from clinical specimens. An over-the-counter test to detect SARS-CoV-2 from clinical specimens is an in vitro diagnostic device for the detection of SARS-CoV-2 in clinical specimens to aid in the diagnosis of SARS-CoV-2 infection. The device is intended to be used by lay users and without required health care provider (HCP) intervention in home settings or similar environments in which lay users perform testing.

Section 513(f)(2) of the Federal Food, Drug, and Cosmetic Act (the FD&C Act) was amended by section 607 of the Food and Drug Administration Safety and Innovation Act (FDASIA) on July 9, 2012. This law provides two options for De Novo classification. First, any person who receives a "not substantially equivalent" (NSE) determination in response to a 510(k) for a device that has not been previously classified under the Act may request FDA to make a risk-based classification of the device under section 513(a)(1) of the Act. On December 13, 2016, the 21st Century Cures Act removed a requirement that a De Novo request be submitted within 30 days of receiving an NSE determination. Alternatively, any person who determines that there is no legally marketed device upon which to base a determination of substantial equivalence may request FDA to make a risk-based classification of the device under section 513(a)(1) of the Act without first submitting a 510(k). FDA shall, within 120 days of receiving such a request, classify the device. This classification shall be the initial classification of the device. Within 30 days after the issuance of an order classifying the device, FDA must publish a notice in the Federal Register announcing the classification.

On May 2, 2022, FDA received your De Novo request for classification of the Cue COVID-19 Molecular Test. The request was submitted under section 513(f)(2) of the FD&C Act. In order to classify the Cue COVID-19 Molecular Test into class I or II, it is necessary that the proposed class have sufficient regulatory controls to provide reasonable assurance of the safety and effectiveness of the device for its intended use. After review of the information submitted in the De Novo request, FDA has determined that, for the previously stated indications for use, the Cue COVID-19 Molecular Test can be classified into class II with the establishment of special controls. FDA believes that class II special controls provide reasonable assurance of the safety and effectiveness of the device type. The identified risks and mitigation measures associated with the device type are summarized in the following table:

Risks to Health	Mitigation Measures
Risk of false results	Certain labeling information including limitations, device descriptions, performance information, and explanations of procedures as identified in special controls (1), (2), (3), (4), (5). Certain design verification and validation including documentation of device descriptions, certain analytical studies and clinical studies, risk analysis strategies identified in special control (6). Testing of characterized viral samples and labeling information identified in special control (7).
Failure to correctly interpret test results	Certain labeling information including limitations, device descriptions, performance information, and explanations of procedures as identified in special controls (1), (2), (3), (4), (5). Certain design verification and validation including documentation of device descriptions, certain analytical studies and clinical studies, risk analysis strategies identified in special control (6).
Failure to correctly operate the device	Certain labeling information including limitations, device descriptions, performance information, and explanations of procedures as identified in special controls (1), (2), (3), (4), and (5). Certain design verification and validation including documentation of device descriptions, certain analytical studies and clinical studies, risk analysis strategies identified in special control (6).

In combination with the general controls of the FD&C Act, the over-the-counter test to detect SARS-CoV-2 from clinical specimens is subject to the following special controls:

1. The intended use in the labeling required under 21 CFR 809.10 must include a description of the following: analytes the device detects and identifies, the specimen types tested, the results provided to the user, the clinical indications for which the test is to be used, the specific intended population(s), and other conditions of use as appropriate.
2. The intended use of the device must only include indications for testing of respiratory specimens that are appropriate for collection by lay users for which there are performance data that demonstrate lay users can collect specimens without health care provider (HCP) supervision in home settings or similar environments.
3. The labeling required under 21 CFR 809.10(b) must include the following:
 - (i) A statement in the intended use that positive results do not rule out co-infection with other respiratory pathogens;
 - (ii) Summary instructions and information written in appropriate language for the intended user that includes easy to follow step-by-step instructions for sample testing,



- explanation of test results, any warnings and precautions relevant to testing, and FAQs, as required by paragraph (b)(5)(iii);
- (iii) Limiting statements including the following, as applicable:
- (A) For those devices intended for testing in symptomatic subjects, a specification of the number of days post symptom onset validated for use of the device and/or a range in which the performance of the test is known, where applicable;
 - (B) Statements that a negative test result does not preclude the possibility of infection with other pathogens, and that a positive test result does not preclude the possibility of co-infection with additional pathogens;
 - (C) A statement that persons with risk factors for severe disease from respiratory pathogens (e.g., chronic lung or heart disease, compromised immune system, diabetes, and other conditions listed by the CDC) should consult and follow-up with a healthcare provider, who will advise if additional testing or treatment are necessary;
 - (D) A statement that the test is not a substitute for consultation with a healthcare provider and should not be used to determine any treatments without provider supervision. A statement that the healthcare provider will consider additional information such as the patient's personal medical history and symptoms, current disease prevalence in the community, and additional test results if applicable, to help determine what steps are best for diagnosis and treatment if needed;
 - (E) A statement that it is especially important to discuss any test results with a healthcare provider if any of the following occur:
 - The symptoms persist or worsen;
 - The patient has high risk for severe illness based on age or medical condition;
 - The patient has a condition that makes it difficult to use the test (e.g., problems with vision, handling the test components, or understanding test instructions or results); or
 - The patient is performing this test on behalf of a person who has any of the above conditions;
 - (F) A statement that accurate results are dependent on adequate product storage, and adherence to the specimen collection and testing procedures. A statement that failure to follow test procedures can lead to incorrect results;
 - (G) A statement that the test must not be used beyond the expiration date listed on the packaging. A statement that use of expired tests can lead to incorrect results;
 - (H) A statement that false positive test results are more likely when prevalence of SARS-CoV-2 is low in the community; and
 - (I) A statement that includes all of the following: The performance characteristics for SARS-CoV-2 were established when [insert predominant strain, subtype, or variant and timeframe] was dominant. Test accuracy may change as new SARS-CoV-2 viruses emerge. Additional testing with a lab-



based molecular test (e.g., PCR) should be considered in situations where a new virus or variant is suspected.

4. The outer box label required under 21 CFR 809.10(a) must include the following:
 - (i) A description of who may use the test, including the presence of symptoms, the days post symptom onset and age restrictions (as applicable);
 - (ii) A list of the components included with the test;
 - (iii) A list of the components required to run the test, but not provided;
 - (iv) A statement that persons with risk factors for severe disease from respiratory pathogens should consult and follow-up with a healthcare provider.
5. The device's labeling must include a prominent hyperlink to the manufacturer's public website where the manufacturer must make the information, identified in this section, publicly and prominently available. The information must include, written in language appropriate for the intended user:
 - (i) A brief summary of the purpose of the test;
 - (ii) Instructions that describe how to appropriately perform the test, interpret the results, and, if applicable, perform follow up testing. The instructions must include the name and intended use of the test, detailed step-by-step instructions of the sample testing procedures, the result(s) interpretation guidance, warnings and limitation statements, information for troubleshooting, and technical assistance with the device (e.g., Help-line contact information).
 - (iii) Frequently Asked Questions (FAQ): this document must provide technical and educational information (e.g., What does this test do and not do, Who should and should not use this test, and directions to resources for further information on the disease and epidemiology).
 - (iv) Information that demonstrates the performance characteristics established in the studies required under paragraph (b)(6).
6. Design verification and validation must include:
 - (i) A detailed device description, including, but not limited to, device components, and a detailed explanation of the methodology, including viral target(s), identification of target detection reagents (e.g., primers, antibodies), internal controls, and computational path from collected raw data to reported result (e.g., how collected raw signals are converted into a reported signal and result), as applicable to the detection method and device design;
 - (ii) Detailed documentation of data from a prospective multisite clinical study with a design and performance that is appropriate for the intended use of the device, including performance estimates derived from a sufficient number of samples from the intended use population for each claimed specimen type. Results must be obtained from geographically diverse locations, such that the performance of the test device is appropriately representative of all present, circulating strains of the claimed viral analyte(s) at the time of the study and submission. Additionally, the clinical study must include diversity across the intended use population and across the clinical range



- of the claimed viral analyte. The clinical study must be performed in the intended use setting (e.g., at home or a home-like environment). The results obtained with the candidate device must be compared to results obtained using a molecular comparator method that FDA has determined to be appropriate. Detailed documentation must include the clinical study protocol (including a predefined statistical analysis plan), study report, testing results, and results of all statistical analyses;
- (iii) The clinical study designs, including number of samples tested, must be sufficient such that the lower bound of the two-sided 95% confidence interval of the positive percent agreement with the comparator must be greater than 70% and additional and appropriate risk mitigation measures are established (e.g., presumptive negative results, serial testing).
 - (iv) Detailed documentation of analytical studies, including those demonstrating the limit of detection (LoD), inclusivity (including relevant variants), cross-reactivity, microbial interference, interfering substances, competitive inhibition, specimen stability, within-lab precision, hook effect, carryover, cross contamination, as applicable;
 - (v) Detailed documentation and characterization (e.g., determination of the identity, supplier, purity, and stability) of all critical reagents and protocols for maintaining product integrity throughout its labeled shelf-life, i.e., reagent stability studies. Data and protocols, including acceptance criteria, from a multi-lot reagent stability study must include testing of samples with challenging analyte concentration, be provided as part of the regulatory submission and must include in-use/open-kit stability, shipping stability, and freeze-thaw stability (as applicable);
 - (vi) Risk analysis and documentation demonstrating how risk control measures are implemented to address device system hazards, such as Failure Modes Effects Analysis and/or Hazard Analysis.
 - (A) This documentation must include a detailed description of a protocol (including all procedures and methods) for the continuous monitoring, identification, and handling of genetic mutations and/or novel isolates or strains (e.g., regular review of published literature and periodic in silico analysis of target sequences to detect possible mismatches). Protocols must include plans to update labeling with additional performance data. All results of this protocol, including any findings, must be documented and must include any additional data analysis that is requested by FDA in response to any performance concerns identified under this section or identified by FDA during routine evaluation. Additionally, if requested by FDA, these evaluations must be submitted to FDA for FDA review within 48 hours of the request and any results that are reasonably interpreted to support the conclusion that novel SARS-CoV-2 strains or isolates impact the stated expected performance of the device must be sent to FDA immediately to the email provided in FDA's request;
 - (B) This must include detailed documentation that demonstrates the effectiveness of risk control measures and device robustness, including the entire testing procedure from sampling to result interpretation, based



on results from the following studies, as applicable per the intended use of the test device: usability studies, user label comprehension, and flex studies;

- (vii) For devices with associated software or instrumentation, documentation must include a detailed description of device software, including software applications and hardware-based devices that incorporate software. The detailed description must include documentation of verification, validation, and hazard analysis and risk assessment activities, including an assessment of the impact of threats and vulnerabilities on device functionality and end users/patients as part of cybersecurity review; and
 - (viii) For devices intended for the detection of SARS-CoV-2 for which an FDA recommended reference material and/or test panel is available, the performance results of an analytical study testing the FDA recommended reference material. Detailed documentation must be kept of that study and its results, including the study protocol, study report for the proposed intended use, testing results, and results of all statistical analyses.
7. If one of the actions listed in section 564(b)(1)(A)–(D) of the Federal Food, Drug, and Cosmetic Act occurs with respect to one or more of the analytes claimed in the intended use, or if the Secretary of Health and Human Services (HHS) determines, under section 319(a) of the Public Health Service Act, that a disease or disorder presents a public health emergency, or that a public health emergency otherwise exists, with respect to SARS-CoV-2:
- (i) Within 30 days from the date that FDA notifies manufacturers that characterized samples are available for test evaluation, the manufacturer must have testing performed on the device with those samples in accordance with a standardized protocol considered and determined by FDA to be acceptable and appropriate; and
 - (ii) Within 60 days from the date that FDA notifies manufacturers that characterized samples are available for test evaluation and continuing until 3 years from that date, the results of the emergency analytical reactivity testing, including the detailed information for the samples tested as described in the certificate of authentication, must be included in a tabular format on the hyperlink the manufacturer’s public website as described in paragraph (b)(5).

Although this letter refers to your product as a device, please be aware that some granted products may instead be combination products. If you have questions on whether your product is a combination product, contact CDRHProductJurisdiction@fda.hhs.gov.

Section 510(m) of the FD&C Act provides that FDA may exempt a class II device from the premarket notification requirements under section 510(k) of the FD&C Act, if FDA determines that premarket notification is not necessary to provide reasonable assurance of the safety and effectiveness of the device type. FDA has determined premarket notification is necessary to provide reasonable assurance of the safety and effectiveness of the device type and, therefore, the device is not exempt from the premarket notification requirements of the FD&C Act. Thus,



persons who intend to market this device type must submit a premarket notification containing information on the over-the-counter test to detect SARS-CoV-2 from clinical specimens.

Please be advised that FDA's decision to grant this De Novo request does not mean that FDA has made a determination that your device complies with other requirements of the FD&C Act or any Federal statutes and regulations administered by other Federal agencies. You must comply with all the FD&C Act's requirements, including, but not limited to: registration and listing (21 CFR Part 807); labeling (21 CFR Parts 801 and 809); medical device reporting (reporting of medical device-related adverse events) (21 CFR 803) for devices or postmarketing safety reporting (21 CFR 4, Subpart B) for combination products (see <https://www.fda.gov/combination-products/guidance-regulatory-information/postmarketing-safety-reportingcombination-products>); good manufacturing practice requirements as set forth in the quality systems (QS) regulation (21 CFR Part 820) for devices or current good manufacturing practices (21 CFR 4, Subpart A) for combination products; and if applicable, the electronic product radiation control provisions (Sections 531-542 of the FD&C Act; 21 CFR 1000-1050).

A notice announcing this classification order will be published in the Federal Register. A copy of this order and supporting documentation are on file in the Dockets Management Branch (HFA-305), Food and Drug Administration, 5630 Fishers Lane, Room 1061, Rockville, MD 20852 and are available for inspection between 9 a.m. and 4 p.m., Monday through Friday.

As a result of this order, you may immediately market your device as described in the De Novo request, subject to the general control provisions of the FD&C Act and the special controls identified in this order.

For comprehensive regulatory information about medical devices and radiation-emitting products, please see Device Advice (<https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance>) and CDRH Learn (<https://www.fda.gov/training-and-continuing-education/cdrh-learn>). Additionally, you may contact the Division of Industry and Consumer Education (DICE) to ask a question about a specific regulatory topic. See the DICE website (<https://www.fda.gov/medical-devices/device-advice-comprehensiveregulatory-assistance/contact-us-division-industry-and-consumer-education-dice>) for more information or contact DICE by email (DICE@fda.hhs.gov) or phone (1-800-638-2041 or 301-796-7100).

If you have any questions concerning the contents of the letter, please contact John McManus at john.mcmanus@fda.hhs.gov.

Sincerely,

Uwe Scherf, M.Sc., Ph.D.
Director, Division of Microbiology Devices
OHT7: Office of In Vitro Diagnostics
Office of Product Evaluation and Quality
Center for Devices and Radiological Health

FDA NEWS RELEASE

FDA Permits Marketing of First COVID-19 At-Home Test Using Traditional Premarket Review Process

[More Press Announcements \(/news-events/newsroom/press-announcements\)](/news-events/newsroom/press-announcements)**For Immediate Release:**

June 06, 2023

[Español \(/news-events/press-announcements/la-fda-permite-la-comercializacion-de-la-primer-prueba-de-covid-19-para-el-hogar-mediante-el\)](/news-events/press-announcements/la-fda-permite-la-comercializacion-de-la-primer-prueba-de-covid-19-para-el-hogar-mediante-el)

Today, the U.S. Food and Drug Administration granted marketing authorization for the Cue COVID-19 Molecular Test. The product is a molecular nucleic acid amplification test (NAAT) that is intended to detect genetic material from SARS-CoV-2 virus present in nasal swabs from adults with signs and symptoms of upper respiratory infection. This test is the first at-home over-the-counter (OTC) test for COVID-19 to be granted marketing authorization using a traditional premarket review pathway and the first ever at-home test authorized using a traditional premarket review pathway for any respiratory illness.

“Today’s authorization is the first at-home COVID-19 test granted marketing authorization outside of emergency use authorities, representing a new era of consumer access to diagnostic tests that can be performed entirely at home,” said Jeff Shuren, M.D., J.D., director of the FDA’s Center for Devices and Radiological Health. “The FDA continues to proactively work with test developers that desire to market their products beyond emergency use authorities. This is part of the FDA’s broader effort to advance the development and availability of at-home tests for a variety of medical conditions to expand patient access to testing.”

The Cue COVID-19 Molecular Test consists of a single-use Cue COVID-19 test cartridge, a single-use Cue sample wand (nasal swab), and the Cue cartridge reader (sold separately). The test also uses the Cue Health app, which displays results when the test is complete. The reusable, battery-operated Cue Cartridge Reader runs the Cue Test Cartridge and communicates results directly to the app in about 20 minutes. In a study reviewed by the FDA, this test correctly identified 98.7% of negative and 92.9% of positive samples in individuals with signs and symptoms of upper respiratory infection.

As with all rapid tests, there is a risk of false positive and false negative results. Individuals who test positive for SARS-CoV-2 should take appropriate precautions to avoid spreading the virus and should seek follow-up care with their physician or healthcare provider as additional testing may be necessary. Negative results should be confirmed by a lab-based molecular test if necessary for patient management. Individuals who test negative and continue to experience symptoms of fever, cough and/or shortness of breath may still have SARS-CoV-2 or another respiratory infection and should seek follow up care with their healthcare provider.

Along with this De Novo authorization, the FDA is establishing criteria called special controls that define the requirements related to labeling and performance testing. When met, the special controls, in combination with general controls, provide a reasonable assurance of safety and effectiveness for tests of this type. This action also creates a new regulatory classification, which means that subsequent devices of the same type with the same intended use may go through the FDA’s 510(k) pathway, whereby devices can obtain clearance by demonstrating substantial equivalence to a predicate device.

The FDA granted the marketing authorization to Cue Health Inc.

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The FDA, an agency within the U.S. Department of Health and Human Services, protects the public health by assuring the safety, effectiveness, and security of human and veterinary drugs, vaccines and other biological products for human use, and

medical devices. The agency also is responsible for the safety and security of our nation's food supply, cosmetics, dietary supplements, products that give off electronic radiation, and for regulating tobacco products.

Inquiries

Media:

✉ [Jim McKinney \(mailto:james.mckinney@fda.hhs.gov\)](mailto:james.mckinney@fda.hhs.gov)

☎ 240-328-7305

Consumer:

☎ 888-INFO-FDA



**510(k) SUBSTANTIAL EQUIVALENCE DETERMINATION
DECISION SUMMARY
ASSAY ONLY**

I Background Information:

A 510(k) Number

K230828

B Applicant

ACON Laboratories, Inc.

C Proprietary and Established Names

Flowflex COVID-19 Antigen Home Test

D Regulatory Information

Table 1: Regulatory information

Product Code(s)	Classification	Regulation Section	Panel
QYT	Class II	21 CFR 866.3984 - Over-The-Counter Test To Detect SARS-Cov-2 From Clinical Specimens	Microbiology

II Submission/Device Overview:

A Purpose For Submission:

To obtain 510(k) clearance for the Flowflex COVID-19 Antigen Home Test. The identical test kit is currently marketed as *Flowflex COVID-19 Antigen Home Test* under Emergency Use Authorization, EUA210494.

B Measurand

Nucleocapsid protein antigen from SARS-Coronavirus 2 (SARS-CoV-2)

C Type of Test

Qualitative lateral flow immunoassay

III Intended Use/Indications for Use:

A Intended Use(s):

See Indications for Use below.

B Indication(s) for Use:

The Flowflex COVID-19 Antigen Home Test is a visually read lateral flow immunoassay device intended for the rapid, qualitative detection of SARS-CoV-2 virus nucleocapsid protein antigen directly in anterior nasal swab specimens from individuals with signs and symptoms of COVID-19 within the first 6 days of symptom onset.

This test is for non-prescription home use by individuals aged 14 years or older testing themselves, or adults testing individuals aged 2 years or older.

The Flowflex COVID-19 Antigen Home Test does not differentiate between SARS-CoV and SARS-CoV-2.

All negative results are presumptive. Symptomatic individuals with an initial negative test result must be re-tested once between 48 and 72 hours after the first test using either an antigen test or a molecular test for SARS-CoV-2. Negative results do not preclude SARS-CoV-2 infections or other pathogens and should not be used as the sole basis for treatment.

Positive results do not rule out co-infection with other respiratory pathogens.

This test is not a substitute for visits to a healthcare provider or appropriate follow-up and should not be used to determine any treatments without provider supervision. Individuals who test negative and experience continued or worsening COVID-19 like symptoms, such as fever, cough and/or shortness of breath, should seek follow up care from their healthcare provider.

The performance characteristics for SARS-CoV-2 were established from December 2022 to March 2023 when SARS-CoV-2 Omicron was dominant. Test accuracy may change as new SARS-CoV-2 viruses emerge. Additional testing with a lab-based molecular test (e.g., PCR) should be considered in situations where a new virus or variant is suspected.

C Special Conditions for Use Statement(s):

OTC - Over The Counter

D Special Instrument Requirements:

Not applicable.

IV Device/System Characteristics:

A Device Description:

The Flowflex COVID-19 Antigen Home Test is a lateral flow immunoassay device intended for the qualitative detection of the nucleocapsid protein antigen from SARS-CoV-2 virus. This test does not differentiate between SARS-CoV and SARS-CoV-2.

The test package is composed of the following components:

- Test Cassettes
- Extraction Buffer Tubes
- Tube holder
- Disposable Nasal Swabs
- Package Insert

The test cassette is assembled with a test strip in a plastic housing, that contains a nitrocellulose membrane with two test lines: a test line (T line) and a control line (C line).

B Principle of Operation:

To perform the test, a nasal swab is collected by the lay user. Swabs can be self-collected by the user (age ≥ 14 years) or collected from another individual by an adult (age ≥ 2 years). For sample processing, the swab is first inserted into the extraction buffer during which the extraction buffer disrupts the virus particles in the specimen, exposing internal viral nucleocapsid antigens. The lysed specimen is then added to the sample well of the test cassette. When an adequate volume of the specimen is added to the sample well (S) of the test cassette, the specimen migrates by capillary action from the sample well over the reagent label pad and across the nitrocellulose membrane test strip. During that migration the reagents in the label pad are solubilized. If SARS-CoV-2 antigens are present in the sample, the antigens bind to the specific anti-SARS-CoV-2 antibody conjugates on the label pad and the antigen/antibody complexes then form sandwich complexes at the test line region (T) of the test strip generating a visible purple colored test line. Unbound conjugate continues to migrate across the nitrocellulose membrane and are captured at the control line region where the complexes will result in a purple control line that indicates adequate operations and sample flow during the test. If no SARS-CoV-2 antigens are present in the sample, the conjugate will only be captured at the control line of the test.

Results are interpreted between 15 and 30 minutes after adding the swab solution into the sample well. A false negative or false positive result may occur if the test result is read before 15 minutes or after 30 minutes.

C Instrument Description Information:

1. Instrument Name:

This is a manually performed visually read Lateral Flow test without an instrument.

2. Specimen Identification:

N/A

3. Specimen Sampling and Handling:

N/A

4. Calibration:
N/A

5. Quality Control:
N/A

I Substantial Equivalence Information:

A Predicate Device Name(s):

Cue COVID-19 Molecular Test

B Predicate 510(k) Number(s):

DEN220028

C Comparison with Predicate(s):

Table 2: Comparison with the predicate device

Device & Predicate Device(s):	DEN220028	K230828
Device Trade Name	Cue COVID-19 Molecular Test	Flowflex COVID-19 Antigen Home Test
Similarities		
Indications For Use	Cue COVID-19 Molecular Test is a qualitative in vitro diagnostic device for the detection of SARS-CoV-2 directly in anterior nasal swab specimens from individuals with signs and symptoms of COVID-19 (i.e., symptomatic). The device is intended to be used by lay users and without required health care provider (HCP) intervention in home settings or similar environments in which lay users perform testing.	The Flowflex COVID-19 Antigen Home Test is a qualitative in vitro diagnostic device for the detection of SARS-CoV-2 in directly in anterior nasal swab specimens from individuals with symptoms of COVID-19 (i.e., symptomatic). The device is intended to be used by lay users and without required health care provider (HCP) intervention in home settings or similar environments in which lay users perform testing.
Intended Use	The Cue COVID-19 Molecular Test is a nucleic acid amplification assay that is used with the Cue Health Monitoring System (Cue Cartridge Reader) for the rapid, qualitative detection of SARS-CoV-2 nucleic acid directly in anterior nasal swab specimens	The Flowflex COVID-19 Antigen Home Test is a visually read lateral flow immunoassay device intended for the rapid, qualitative detection of SARS-CoV-2 virus nucleocapsid protein antigen directly in anterior nasal swab specimens from individuals with signs and

	from individuals with signs and symptoms of COVID-19 (i.e., symptomatic). A negative test result is presumptive, and it is recommended these results be confirmed by a lab-based molecular SARS-CoV-2 assay if necessary for patient management. Negative results do not preclude SARS-CoV-2 infections and should not be used as the sole basis for treatment. Positive results do not rule out co-infection with other respiratory pathogens. This test is not a substitute for visits to a healthcare provider or appropriate follow-up and should not be used to determine any treatments without provider supervision. This test is intended to be sold over-the-counter (OTC) for testing of individuals 18 years of age and older.	symptoms of COVID-19 within the first 6 days of symptom onset. This test is for non-prescription home use by individuals aged 14 years or older testing themselves, or adults testing individuals aged 2 years or older. The Flowflex COVID-19 Antigen Home Test does not differentiate between SARS-CoV and SARS-CoV-2. All negative results are presumptive. Symptomatic individuals with an initial negative test result must be re-tested once between 48 and 72 hours after the first test using either an antigen test or a molecular test for SARS-CoV-2. Negative results do not preclude SARS-CoV-2 infections or other pathogens and should not be used as the sole basis for treatment. Positive results do not rule out co-infection with other respiratory pathogens. Performance characteristics for SARS-CoV-2 were established from December 2022 to March 2023 of the SARS-CoV-2 pandemic when SARS-CoV-2 Omicron was the predominant SARS-CoV-2 variant in circulation. When other SARS-CoV-2 virus variant are emerging, performance characteristics may vary. This test is not a substitute for visits to a healthcare provider or appropriate follow-up and should not be used to determine any treatments without provider supervision. Individuals who test negative and experience continued or worsening COVID-19 like symptoms, such as fever, cough and/or shortness of breath, should seek follow up care from their healthcare provider.
Regulation number	21 CFR 866.3984	Same

Disease	COVID-19	Same
Intended Use Population	Individuals with symptoms of COVID-19	Same
Patient Use	Over the counter use/self-testing	Same
Usage	Single use test	Same
Test Result	Qualitative	Same
Format	Test cartridge	Same
Sample Type	Direct anterior swab specimen	Same
Differences		
Analyte	SARS-CoV-2 nucleocapsid nucleic acid	SARS-CoV-2 nucleocapsid protein
Test Principle	Nucleic acid amplification assay	Lateral flow immunoassay
Instrument	Yes	No
Development Time	20 min	15-30 min
Result Interpretation	Instrument read	Visually read
Utilizes App to Display Test Results	Yes	No

The Flowflex COVID-19 Antigen Home Test (proposed device) employs a lateral flow immunoassay design requiring a visual read of test strip for manual interpretation of the test results by the user. In contrast, the Cue COVID-19 Molecular Test (predicate device) is a molecular nucleic acid amplification test. Despite the technological differences, both the proposed candidate device and the predicate device are both over-the-counter test for the detection of SARS-CoV-2 in clinical specimens to aid in the diagnosis of SARS-CoV-2 infection. Accordingly they are grouped into the same Classification under 21 CFR 866.3984. The devices are intended to be used by lay users and do not require a health care provider (HCP) intervention. A Benefit/Risk analysis was provided to address the technological differences of the investigational device when compared to the predicates.

Consistent with the FDA Guidance Document [“Benefit-Risk Factors to Consider When Determining Substantial Equivalence in Premarket Notifications \(510\(k\)\) with Different Technological Characteristics | FDA.”](#) the Agency determined that these differences of the investigations device compared to the predicate do not affect the overall substantial equivalence of the proposed device to the predicate device in terms of intended use, safety and effectiveness.

II Standards/Guidance Documents Referenced:

Document	Title	Publisher	Applicable Study
Nonbinding recommendations for premarket authorization of SARS CoV-2 antigen tests	Premarket validation recommendations for developers of in vitro diagnostic tests for SARS-CoV-2 antigen	FDA/CDRH/OHT7/DMD	All Studies
DEN220028/ Special controls for Over- the-counter test to detect SARS-CoV-2 from clinical specimens 21 CFR866.3984	Reclassification order for DEN220028 and special controls under 21 CFR 866.3984	FDA/CDRH	All Studies
CLIA Waiver FDA Guidance	Recommendations for Clinical Laboratory Improvement Amendments of 1988 (CLIA) Waiver Applications for Manufacturers of In Vitro Diagnostic Devices - Guidance for Industry and Food and Drug Administration Staff (fda.gov)	FDA/CDRH	Flex Studies
ISO11135:2014	Sterilization of health care products - Ethylene oxide - Requirements for development, validation and routine control of a sterilization process for medical devices	ISO	Sterility
ISO 10993-7	Biological Evaluation of Medical Devices – Part 7: Ethylene Oxide Sterilization Residuals	ISO	Sterility
FDA Guidance	Submission and Review of Sterility Information in Premarket Notification (510(k)) Submissions for Devices Labeled as Sterile FDA.	FDA/CDRH	Sterility
ISO 10993-5	Biological evaluation of medical devices Part 5: Tests for in vitro cytotoxicity	ISO	Biocompatibility
ISO 10993-10	Biological evaluation of medical devices Part 10: Tests for irritation and skin sensitization	ISO	Biocompatibility

III Performance Characteristics (if/when applicable):

A Analytical Performance:

1. Multi-lot Precision:

The purpose of the study was to assess lot-to-lot variability of three different lots of the Flowflex COVID-19 Antigen Home Test. Three (3) levels of heat-inactivated SARS-CoV-2 were spiked in surrogate clinical matrix (1xPBS with 1% mucin) as follows:

- A. Negative, Surrogate Negative Matrix
- B. Low positive (1.5x LoD)
- C. Positive (5x LoD)

A matrix comparison study was conducted in support of the surrogate matrix and demonstrated the equivalency of the surrogate matrix to negative clinical nasal swab matrix.

Each operator applied 50 µL of each coded sample to each dry nasal swab. Then, the operator processed the sample per the IFU of the proposed device. The sample panel was tested in a blinded manner by three operators for 10 non-consecutive days. All three lots were tested by each operator on each testing day. Each sample level was tested in duplicate in each of two runs per operator per day (i.e., 3 lots x 3 operators x 2 replicates/run x 2 runs X 10 days). A total of 360 test were run per panel member.

The agreement of obtained results with expected results was 100% across all lots, operators, and days. Variability in results was not observed between the three independently manufactured lots.

The results are summarized below:

Table 2: Multi-lot precision study results performed by 3 operators for 10 days

Lot #	Negative*	Low positive** (1xLoD)	Positive** (5xLoD)
COV3030007 (lot 1)	120/120	120/120	120/120
COV3030009 (lot 2)	120/120	120/120	120/120
MPS2100029 (lot 3)	120/120	120/120	120/120
% Agreement (95% CI***)	100% [99%-100%]	100% [99%-100%]	100% [99%-100%]

*Virus not detected/total; **Virus detected/total; ***CI: Confidence Interval

2. Linearity:

Not applicable, the device is a binary qualitative assay that is visually read.

3. Analytical Specificity/Interference:

a) *Cross Reactivity/Microbial Interference:*

The analytical specificity of the Flowflex COVID-19 Antigen Home Test was evaluated by testing various microorganisms (13), and viruses (17) diluted into negative clinical nasal swab matrix (see LoD study section below for generation of the negative clinical nasal swab matrix). Nasal wash was tested as representative of normal respiratory microbial flora. Each organism and virus were tested in three replicates in the absence (cross reactivity) or presence (microbial interference) of 4.57×10^5 TCID₅₀/mL (3x LoD) of heat-inactivated SARS-CoV-2 (isolate USA-WA1/2020). The microbial interference and the cross-reactivity study were conducted simultaneously with samples tested in a randomized and blinded manner.

To demonstrate that the Flowflex COVID-19 Antigen Home Test does not react with related viruses, high prevalence disease agents, and normal or pathogenic flora that are reasonably likely to be encountered in nasal swab specimens were tested. Viruses were tested at concentrations of $\geq 10^5$ TCID₅₀/mL (PFU/mL or copies/mL), Bacteria and fungi were tested at concentrations of $\geq 10^6$ CFU/mL. The results are summarized in the table below:

Table 3: Summary of cross-reactivity and microbial interference study results

Microorganisms	Without SARS-CoV-2 (Cross-Reactivity)		With SARS-CoV-2 (Microbial Interference)	
	Concentration Tested*	Positive/ Tested	Concentration Tested*	Positive/ Tested
Adenovirus	2.82 x 10 ⁶ TCID ₅₀ /mL	0/3	2.82 x 10 ⁶ TCID ₅₀ /mL	3/3
<i>Bordetella pertussis</i>	1.16 x 10 ⁹ CFU/mL	0/3	1.16 x 10 ⁹ CFU/mL	3/3
<i>Candida albicans</i>	1.31 x 10 ⁷ CFU/mL	0/3	1.31 x 10 ⁷ CFU/mL	3/3
<i>Chlamydia pneumonia</i>	1.4 x 10 ⁷ IFU/mL	0/3	1.4 x 10 ⁷ IFU/mL	3/3
<i>Chlamydia trachomatis</i>	3.52 x 10 ⁸ IFU/mL	0/3	3.52 x 10 ⁸ IFU/mL	3/3
Enterovirus	1.0 x 10 ⁵ TCID ₅₀ /mL	0/3	1.0 x 10 ⁵ TCID ₅₀ /mL	3/3
Haemophilus influenzae	3.87 x 10 ⁷ CFU/mL	0/3	3.87 x 10 ⁷ CFU/mL	3/3
Human coronavirus 229E	1.0 x 10 ⁵ TCID ₅₀ /mL	0/3	1.0 x 10 ⁵ TCID ₅₀ /mL	3/3
Human coronavirus HKU1 (specimen # B18378425)	Ct 13.7	0/3	Ct 13.7	3/3
Human coronavirus HKU1 (specimen # B18431603)	Ct 20.3	0/3	Ct 20.3	3/3
Human coronavirus HKU1 (specimen # B18427097)	Ct 24.6	0/3	Ct 24.6	3/3
Human coronavirus HKU1 (specimen # B18431601)	Ct 21.3	0/3	Ct 21.3	3/3
Human coronavirus HKU1 (specimen # B18438368)	Ct 16.8	0/3	Ct 16.8	3/3
Human coronavirus HKU1 (specimen # B18421181)	Ct 19.8	0/3	Ct 19.8	3/3
Human coronavirus NL63	1.0 x 10 ⁵ TCID ₅₀ /mL	0/3	1.0 x 10 ⁵ TCID ₅₀ /mL	3/3
Human coronavirus OC43	1.0 x 10 ⁵ TCID ₅₀ /mL	0/3	1.0 x 10 ⁵ TCID ₅₀ /mL	3/3
Human Metapneumovirus	1.0 x 10 ⁵ TCID ₅₀ /mL	0/3	1.0 x 10 ⁵ TCID ₅₀ /mL	3/3
Influenza A	1.51 x 10 ⁵ TCID ₅₀ /mL	0/3	1.51 x 10 ⁵ TCID ₅₀ /mL	3/3
Influenza B	1.0 x 10 ⁵ TCID ₅₀ /mL	0/3	1.0 x 10 ⁵ TCID ₅₀ /mL	3/3
<i>Legionella pneumophila</i>	3.27 x 10 ⁹ CFU/mL	0/3	3.27 x 10 ⁹ CFU/mL	3/3
MERS-coronavirus	1.05 x 10 ⁵ TCID ₅₀ /mL	0/3	1.05 x 10 ⁵ TCID ₅₀ /mL	3/3

<i>Mycobacterium tuberculosis</i>	1.21 x 10 ⁷ CFU/mL	0/3	1.21 x 10 ⁷ CFU/mL	3/3
<i>Mycoplasma pneumoniae</i>	2.70 x 10 ⁷ CCU/mL	0/3	2.70 x 10 ⁷ CCU/mL	3/3
Parainfluenza virus 1	3.80 x 10 ⁵ TCID ₅₀ /mL	0/3	3.80 x 10 ⁵ TCID ₅₀ /mL	3/3
Parainfluenza virus 2	3.39 x 10 ⁶ TCID ₅₀ /mL	0/3	3.39 x 10 ⁶ TCID ₅₀ /mL	3/3
Parainfluenza virus 3	1.15 x 10 ⁶ TCID ₅₀ /mL	0/3	1.15 x 10 ⁶ TCID ₅₀ /mL	3/3
Parainfluenza virus 4	9.55 x 10 ⁵ TCID ₅₀ /mL	0/3	9.55 x 10 ⁵ TCID ₅₀ /mL	3/3
Pooled human nasal cavity wash	n/a	0/3	n/a	3/3
<i>Pseudomonas aeruginosa</i>	4.93 x 10 ⁸ CFU/mL	0/3	4.93 x 10 ⁸ CFU/mL	3/3
Respiratory syncytial virus	1.0 x 10 ⁵ TCID ₅₀ /mL	0/3	1.0 x 10 ⁵ TCID ₅₀ /mL	3/3
Rhinovirus	1.0 x 10 ⁵ TCID ₅₀ /mL	0/3	1.0 x 10 ⁵ TCID ₅₀ /mL	3/3
<i>Staphylococcus aureus</i>	9.23 x 10 ⁷ CFU/mL	0/3	9.23 x 10 ⁷ CFU/mL	3/3
<i>Staphylococcus epidermidis</i>	6.84 x 10 ⁸ CFU/mL	0/3	6.84 x 10 ⁸ CFU/mL	3/3
<i>Streptococcus pneumoniae</i>	7.22 x 10 ⁷ CFU/mL	0/3	7.22 x 10 ⁷ CFU/mL	3/3
<i>Streptococcus pyogenes</i>	4.60 x 10 ⁷ CFU/mL	0/3	4.60 x 10 ⁷ CFU/mL	3/3

None of the organisms and viruses listed above showed cross-reactivity and interference in the assay at the concentrations listed.

The Sponsor performed *in silico* analysis for SARS-Coronavirus instead of the recommended wet-testing as the virus was not available. Cross reactivity of the test with SARS-Coronavirus could not be excluded from the *in silico* analysis. Accordingly, the intended use of the proposed device states that it cannot distinguish SARS-CoV-2 and SARS-CoV, the *in silico* analysis is acceptable.

The *in silico* analysis of *Pneumocystis jirovecii* identified two regions on the nucleocapsid protein of SARS-CoV-2 (P0DTC9), between amino acids 4-47 and 109-171. Both regions showed very low similarity compared to the translated amino acid sequences of *Pneumocystis jirovecii*, 31.8% and 28.1% percent identity, respectively. The longest consecutive matches had only 3 amino acids. Due to the low homology between the SARS-CoV-2 nucleocapsid protein and the amino acid sequences of *Pneumocystis jirovecii*, cross-reactivity or microbial interference of *Pneumocystis jirovecii* with the Flowflex COVID-19 Antigen Home Test is highly unlikely.

b) Interfering Substances:

Thirty-one (31) potentially interfering substances, diluted in negative clinical nasal swab matrix (see LoD study section below for generation of the negative clinical nasal swab matrix), were evaluated with the Flowflex COVID-19 Antigen Home Test. Each substance was tested in three (3) replicates in the absence or presence of heat-inactivated SARS-CoV-2 (isolate USA-WA1/2020) at 3x LoD (4.57×10^3 TCID₅₀/mL). Based on the test results, the endogenous and exogenous interfering substances tested at a concentration listed in Table 3 do not cross-react or interfere with the performance of the Flowflex COVID-19 Antigen Home Test at the concentrations tested.

Table 4: Interfering Substances Study Results

Interfering Substances Tested	No SARS-CoV-2		SARS-CoV-2 virus (3x LoD)	
	Concentration Tested	Positive/ Tested	Concentration Tested	Positive/ Tested
Beclomethasone	5 mg/mL	0/3	5 mg/mL	3/3
Biotin	3500 ng/mL	0/3	3,500 ng/mL	3/3
Dexamethasone	10 mg/mL	0/3	10 mg/mL	3/3
Dyclonine Hydrochloride	2 mg/mL	0/3	2 mg/mL	3/3
Flunisolide	10 mg/mL	0/3	10 mg/mL	3/3
Hand sanitizer	15% v/v	0/3	15% v/v	3/3
Hand Soap	15% v/v	0/3	15% v/v	3/3
Homeopathic allergy relief (histaminum hydrochloricum)	15% w/v	0/3	15% w/v	3/3
Homeopathic nasal wash (alkalol)	5% v/v	0/3	5% v/v	3/3
Leukocytes	4.8×10^6 cells/mL	0/3	4.8×10^6 cells/mL	3/3
Molnupiravir	10 mg/mL	0/3	10 mg/mL	3/3
Mucin	2.5 mg/mL	0/3	2.5 mg/mL	3/3
Mupirocin	10 mg/mL	0/3	10 mg/mL	3/3
Nasal corticosteroids (Budesonide)	15% v/v	0/3	15% v/v	3/3
Nasal corticosteroids (fluticasone furate)	5% v/v	0/3	5% v/v	3/3
Nasal corticosteroids (fluticasone propionate)	5% v/v	0/3	5% v/v	3/3
Nasal corticosteroids (Mometasone furoate)	15% v/v	0/3	15% v/v	3/3
Nasal corticosteroids	15% v/v	0/3	15% v/v	3/3

Interfering Substances Tested	No SARS-CoV-2		SARS-CoV-2 virus (3x LoD)	
	Concentration Tested	Positive/ Tested	Concentration Tested	Positive/ Tested
(Triamcinolone Acetonide)				
Nasal decongestant (Galphimia glauca, Luffa operculata, sabadilla)	15% v/v	0/3	15% v/v	3/3
Nasal gel	5% v/v	0/3	5% v/v	3/3
Nasal spray (Cromolyn sodium nasal solution)	15% v/v	0/3	15% v/v	3/3
Nasal spray (Oxymetazoline HCl)	15% v/v	0/3	15% v/v	3/3
Nasal spray (Phenylephrine HCl)	15% v/v	0/3	15% v/v	3/3
Nasal spray (Sodium Chloride & Preservatives)	15% v/v	0/3	15% v/v	3/3
Oral Anesthetic Cough Lozenge (Menthol)	3 mg/mL	0/3	3 mg/mL	3/3
Oseltamivir Phosphate (Tamiflu)	15% w/v	0/3	15% w/v	3/3
Remdesivir	10 mg/mL	0/3	10 mg/mL	3/3
Sore Throat & Cough Lozenges (Benzocaine, Dextromethorphan HBr)	3 mg/mL	0/3	3 mg/mL	3/3
Sore Throat Spray (Phenol)	5% v/v	0/3	5% v/v	3/3
Tobramycin	50 ug/mL	0/3	50 ug/mL	3/3
Whole Blood	2.5% v/v	0/3	2.5%	3/3

4. Assay Reportable Range:

Not applicable, the device is a binary qualitative assay that is visually read.

5. Traceability, Stability, Expected Values (Controls, Calibrators, or Methods):

a) Internal Controls:

The Flowflex Plus COVID-19 Home Test has a built-in internal procedural control. The colored biotin labeled particle will bind to streptavidin on nitrocellulose membrane and the colored mouse monoclonal antibody labeled particle will bind to goat anti-mouse antibody on nitrocellulose membrane to form a colored Control line (C) in the control line region. Formation

of the Control line serves as an internal control indicating that proper volume of specimen has been added and membrane wicking has occurred.

b) Sample Stability:

Positive nasal swab samples were generated by spiking heat inactivated SARS-CoV-2 WA1/2020 at 3x LoD (4.57×10^5 TCID₅₀/mL) into negative clinical nasal swab matrix and pipetting 50 µL on each swab. Negative swabs (50 µL negative clinical nasal swab matrix) and moderate positive swabs were then stored at 10°C, 15°C, 30°C, 35°C, and ambient temperature (25°C) for 1 hour, 2 hours, 3 hours and 4 hours before swabs were placed into the extraction buffer tubes for sample processing. Freshly prepared swabs were tested immediately as a control condition. The nasal swabs stored at each temperature condition for each time point were prepared in 5 replicates and tested with Flowflex COVID-19 Antigen Home Test per its IFU. The results are summarized in table below:

Table 5: Test results for sample stability study.

Temperature Condition	Time points	Negative sample					Positive sample at 3xLoD (4.57×10^5 TCID ₅₀ /mL)				
		Swab-1	Swab-2	Swab-3	Swab-4	Swab-5	Swab-1	Swab-2	Swab-3	Swab-4	Swab-5
	Freshly prepared	-	-	-	-	-	+	+	+	+	+
10°C	1 hour	-	-	-	-	-	+	+	+	+	+
	2 hours	-	-	-	-	-	+	+	+	+	+
	3 hours	-	-	-	-	-	+	+	+	+	+
	4 hours	-	-	-	-	-	+	+	+	+	+
15°C	1 hour	-	-	-	-	-	+	+	+	+	+
	2 hours	-	-	-	-	-	+	+	+	+	+
	3 hours	-	-	-	-	-	+	+	+	+	+
	4 hours	-	-	-	-	-	+	+	+	+	+
25°C	1 hour	-	-	-	-	-	+	+	+	+	+
	2 hours	-	-	-	-	-	+	+	+	+	+
	3 hours	-	-	-	-	-	+	+	+	+	+
	4 hours	-	-	-	-	-	+	+	+	+	+
30°C	1 hour	-	-	-	-	-	+	+	+	+	+
	2 hours	-	-	-	-	-	+	+	+	+	+
	3 hours	-	-	-	-	-	+	+	+	+	+
	4 hours	-	-	-	-	-	+	+	+	+	+
35°C	1 hour	-	-	-	-	-	+	+	+	+	+
	2 hours	-	-	-	-	-	+	+	+	+	+
	3 hours	-	-	-	-	-	+	+	+	+	+
	4 hours	-	-	-	-	-	+	+	+	+	+

The results read at 15 min were the same results as read at 30 min.

c) Device Stability:

In order to determine the stability of the Flowflex Plus COVID-19 Home Test three (3) test kits were stored at 2-8°C and at 30°C for 27 months. The shelf-life assessment started within one month of test kit manufacturing. Testing supports storage of the test at 2- 30°C for 24 months.

d) Shipping Stability:

In order to understand the stability of the test kit under different shipping conditions, Flowflex COVID-19 Antigen Home Test kits from a single lot were stored at three different temperature conditions: 60°C with 85% ±5% relative humidity, room temperature (25°C) and 2~8°C (as a

reference) for 8 days. The stored test kits were tested daily with five (5) replicates. All negative samples tested negative, all positive samples ($3 \times \text{LoD}$, $4.57 \times 10^3 \text{ TCID}_{50}/\text{mL}$) tested positive for all storage conditions.

In addition, the impact of harsh shipping conditions on the performance of Flowflex COVID-19 Antigen Home Test was assessed for three (3) different test lots that were stored at -20°C for 24 hours and then stored at room temperature for 24 hours. Three (3) freeze/thaw cycles were repeated after which the products were stored at 55°C for 35 days. All negative samples tested negative, all positive samples ($3 \times \text{LoD}$, $4.57 \times 10^3 \text{ TCID}_{50}/\text{mL}$) tested positive in all storage conditions over the 35 days.

6. Detection Limit:

a) Limit of Detection with heat-inactivated SARS-CoV-2 isolate USA-WA1/2020

The Limit of Detection (LoD) of the Flowflex COVID-19 Antigen Home Test was determined by evaluating different dilutions of heat-inactivated SARS-CoV-2 isolate USA-WA1/2020 into pooled negative clinical nasal swab matrix collected from multiple COVID-19 negative individuals. Each nostril was swabbed with a new swab, and each swab was eluted in 0.3 mL of PBS. The swab eluates were combined and mixed thoroughly. The pooled negative clinical nasal swab matrix was confirmed as negative for SARS-CoV-2 by RT-PCR. The LoD was determined as the lowest virus concentration that was detected $\geq 95\%$ of the time (i.e., concentration at which at least 19 out of 20 replicates tested positive). The limit of detection was established in two phases.

Range Finding LoD Study

Five serial dilutions were made from heat inactivated SARS-CoV-2 WA1/2020 virus into negative clinical nasal swab matrix. Five replicates were tested on two kit lots of the assay for each of five 1:10 dilutions to determine the preliminary LoD concentration of the device. The lowest concentration with 5/5 positive results from each lot was considered the preliminary LoD for each virus strain. For each replicate 50 μL of virus dilution was applied to a swab and the swab was processed according to the IFU. The results are summarized below.

Table 6: Preliminary LoD Study Summary for SARS-CoV-2 WA1/2020

Concentration of SARS CoV-2 WA1/2020 applied to dry swab	Kit Lot MPS2100029 Positive/ Tested	Kit Lot MPS3020028 Positive/ Tested
$4.57 \times 10^5 \text{ TCID}_{50}/\text{mL}$	5/5	5/5
$4.57 \times 10^4 \text{ TCID}_{50}/\text{mL}$	5/5	5/5
$4.57 \times 10^3 \text{ TCID}_{50}/\text{mL}$	5/5	5/5
$4.57 \times 10^2 \text{ TCID}_{50}/\text{mL}$	3/5	3/5

The results read at 15 min were the same as results read at 30 min.

Confirmatory LoD Study

The Preliminary LoD concentration and the concentrations above and below the LoD were tested with a total of twenty ($n = 20$) replicates of the same inactivated SARS-CoV-2 material for each of two test kit lots. To confirm the LoD for each lot, at least 19 of the 20 replicates should be positive. Per lot, the final confirmation data set included the confirmed LoD level with at least one additional level tested above and below to demonstrate that levels above LoD were 100% positive and levels below LoD were $<95\%$ positive. The results are summarized below.

Table 7: Confirmatory LoD Study Summary for SARS-CoV-2 WA1/2020

Concentration of SARS CoV-2 WA1/2020 applied to dry swab	Lot MPS2100029 Positive/ Tested	Lot MPS3020028 Positive/ Tested
1.37 x 10 ⁴ TCID ₅₀ /mL (3-fold above the preliminary LoD)	20/20	20/20
4.57 x 10 ³ TCID ₅₀ /mL (Preliminary LoD)	20/20	20/20
1.52 x 10³ TCID₅₀/mL (3-fold below the preliminary LoD)	20/20	20/20
4.57 x 10 ² TCID ₅₀ /mL	2/5	2/5

The limit of detection for the Flowflex COVID-19 Antigen Home Test using SARS-CoV-2 strain WA1/2020, was confirmed to be 1.52 x 10³ TCID₅₀/mL (7.6 x 10 TCID₅₀/swab) with both tested lots. In addition, 3 lots had previously been tested with this device under EUA 210494 using a different lot of SARS-CoV-2 WA1/2020 and resulted in a similar LoD of 2.5 x 10³ TCID₅₀/mL (1.3 x 10² TCID₅₀/swab).

b) 1st WHO International Standard for SARS-CoV-2 antigen (NIBSC code: 21/368)

The LoD of the Flowflex COVID-19 Antigen HomeTest was also determined by evaluating different dilutions of WHO International Standard for SARS-CoV-2 antigen (NIBSC code: 21/368) in negative clinical nasal swab matrix (generated as described in the LoD study section above). Three ampoules of WHO International Standard for SARS-CoV-2 antigen were used for testing two lots of the Flowflex COVID-19 Antigen Home Test. Each ampoule of the lyophilized SARS-CoV-2 antigen standard was reconstituted in 0.25 mL of ultra-pure water and pooled together (concentration 20,000 IU/mL). The LoD was determined as the lowest virus concentration that was detected ≥ 95% of the time (i.e., concentration at which at least 19 out of 20 replicates tested positive). The limit of detection was established in two phases.

Range Finding LoD Study

Five serial dilutions were made from the WHO International Standard for SARS-CoV-2 antigen into negative clinical matrix. Three replicates were tested on two kit lots of the assay for each of five 1:2 dilutions to determine the preliminary LoD concentration of the device. The lowest concentration with 3/3 positive results from each lot was considered the preliminary LoD. For each replicate 50 µL of virus dilution was applied to a swab and the swab was processed according to the IFU. The results are summarized below.

Table 8: Preliminary LoD Study Summary for WHO International Standard for SARS-CoV-2 antigen

Concentration of WHO International Standard for SARS-CoV-2 antigen applied to dry swab	Kit Lot MPS2100029 Positive/ Tested	Kit Lot MPS3020028 Positive/ Tested
10,000 IU/mL	3/3	3/3
5,000 IU/mL	3/3	3/3
2,500 IU/mL	3/3	3/3
1,250 IU/mL	3/3	3/3
625 IU/mL	3/3	3/3
312.5 IU/mL	2/3	1/3

The results read at 15 min were the same as results read at 30 min.

Confirmatory LoD Study

The Preliminary LoD concentration and the concentrations above and below the LoD were tested with a total of twenty (n = 20) replicates for each of two test kit lots. Sponsor tested a total of twenty (n = 20) replicates for the confirmatory dilutions for WHO International Standard for SARS-CoV-2 antigen. To confirm the LoD for each lot, at least 19 of the 20 replicates should be positive. The results are summarized below.

Table 9: Confirmatory LoD Study Summary for WHO International Standard for SARS-CoV-2 antigen

Concentration of WHO International Standard for SARS-CoV-2 antigen applied to dry swab	IU/Swab	Test concentration relative to LoD	Lot MPS2100029 Positive/ Tested	Lot MPS3020028 Positive/ Tested
1875 IU/mL	(94 IU/swab)	3x	20/20	20/20
625 IU/mL	(31 IU/swab)	1x	20/20	20/20
208 IU/mL	(10 IU/swab)	0.3x	8/20	7/20

The LoD for the Flowflex COVID-19 Antigen Home Test using 1st WHO International Standard for SARS-CoV-2 antigen (NIBSC code: 21/368) in nasal matrix was confirmed to be 625 IU/mL (31 IU/swab).

7. High Dose Hook Effect

The purpose of this study was to evaluate the effect of a high concentration of SARS antigen on the performance of the Flowflex COVID-19 Antigen Home Test. Five replicates were tested for each lot. 50 µl of heat-inactivated SARS-CoV-2 (isolate WA1/2020) diluted into negative clinical matrix at 2.82×10^7 TCID₅₀/mL, were added onto each swab and tested on the Flowflex COVID-19 Antigen Home Test. The testing was conducted according to the Instructions for Use for the test. The results are summarized below.

Table 10: Summary of High Dose Hook Effect Study

Concentration of SARS CoV-2 virus applied to dry swab	Lot#COV3030007 (Positive/Tested)	Lot#COV3030009 (Positive/Tested)	Lot#MPS2100029 (Positive/Tested)
2.82×10^7 TCID ₅₀ /mL	5/5	5/5	5/5
A very strong Test Line was observed 10-15 seconds after the addition of sample solution on each test cassette.			

High-dose hook effect was not observed when tested with concentrations up to 2.82×10^7 TCID₅₀/mL of heat-inactivated SARS-CoV-2 virus.

8. Inclusivity (Analytical Reactivity):

Analytical reactivity for Flowflex COVID-19 Antigen Home Test was demonstrated using 6 additional strains/isolates of SARS CoV-2 virus. Heat-inactivated SARS-CoV-2 isolates were each diluted into negative clinical nasal swab matrix at different concentrations (see LoD study section above for generation of the negative clinical nasal swab matrix). Each concentration was tested with 5 replicates until two consecutive dilutions produced one or more negative replicates out of 5. The reactivity of the Flowflex COVID-19 Antigen Home Test with the variants is

summarized below with the lowest concentration that returned 100% positive replicates (i.e., 5/5):

Table 11. Summary of the Flowflex COVID-19 test's reactivity with SARS-CoV-2 variants

SARS-CoV-2 Variants	Lowest Variant Concentration with 5/5 Positive Replicates [TCID ₅₀ /mL]
SARS-CoV-2 virus (USA-WA1/2020)	1.52 x 10 ³
B.1.1.7 (Alpha)	4.20 x 10 ²
B.1.351 (Beta)	1.05 x 10 ³
P.1 (Gamma)	4.20 x 10 ²
B.1.617.2 (Delta)	1.05 x 10 ³
B.1.1.529 (Omicron)	5.01 x 10 ²
BA.2.3 (Omicron)	2.45 x 10 ³

A plan has been established by the sponsor to closely monitor for the emergence of any circulating variants of concern. The plan includes the monitoring of publicly available databases and information from public health authorities, and a multi-tier approach that determines reactivity of the test with emerging variants (including silico analysis, use of antibody escape mutational profile, and wet testing).

9. Assay Cut-Off:

Not applicable, the device is a qualitative assay with a binary result.

A Comparison Studies:

1. Method Comparison with Predicate Device:

Not applicable. See "C. Clinical Studies." for clinical performance.

2. Matrix Comparison:

The Flowflex COVID-19 Antigen Home Test is only intended for the qualitative detection of the nucleocapsid protein antigen from SARS-CoV-2 in direct anterior nasal swab specimens. As no other specimen or sample type is claimed for this device. A Matrix Comparison study to support other sample types for clinical testing with this test was therefore not performed.

However, a matrix equivalency study between negative clinical nasal swab matrix and the surrogate matrix composed of 1% mucin in PBS, that was used in the precision study was conducted and demonstrated equivalent performance of the test with both matrices.

B Clinical Studies:

1. Clinical Sensitivity and Specificity:

The performance of the Flowflex COVID-19 Antigen Home Test was compared to the sample results as generated by a highly sensitive EUA authorized molecular SARS-CoV-2 RT-PCR assay with an extraction step. A prospective clinical study was conducted from December 2022 to February 2023. Nine clinical sites (Sites A- I) participated in the study.

A nasopharyngeal (NP) swab was collected first from symptomatic patients by a healthcare professional at the clinical study sites who inserted the swab into a tube containing 3 mL of Viral Transport Medium for comparator testing.

Thereafter, anterior nasal swab samples for the investigational device were either self-collected by a lay user aged ≥ 14 years or collected by an adult (parent/guardian) from individuals aged 2 - <14 years. Anterior nasal swabs were collected and processed by the lay users per the IFU of the proposed device. A total of 804 subjects were enrolled in this study, 699 subjects (age ≥ 14 years old) and 99 subjects ages 2 - 13 years. Six (6) subjects age ≥ 14 participated in paired-study as a donor and their nasal swab specimens were collected and tested by another person with age >14 . Detailed study subject demographics are given below:

Table 12: Demographics - Age distribution and education of the clinical study cohort

Age group	Percent total	Total	Education	Lay user	Total
2-13 years*	12%	99	Some High School	24**	3%
14-24 years	10%	81	High School	508	63.6%
25-64 years	64%	514	Some College	190	23.8%
≥ 65 years	13%	105	Bachelors	53	6.6%
			Master	18	2.2%
			Doctorate	6	0.8%
Total	NA	799	Total	799	100%

*Tested by adult >14 years old per the intended use statement.

**24 users age 14 to <18 did not finish high school yet

The clinical study included about 22% low positive samples. The Flowflex COVID-19 Antigen Home Test detected SARS-CoV-2 with Positive Percent Agreement (PPA) of 89.8% and Negative Percent Agreement (NPA) of 99.3% when compared to the result of the SARS-CoV-2 RT-PCR comparator assay. Results are listed in the table below.

Table 13: Performance of Flowflex COVID-19 Test compared to high sensitivity RT-PCR COVID-19.

Flowflex COVID-19 Antigen Home Test	EUA RT-PCR Comparator		
	Positive	Negative	Total
Positive	212	4	216
Negative	24	559	583
Total	236	563	799
Positive Percent Agreement (PPA)	89.8% (212/236) (95%CI: 85.2% - 93.4%)		

Negative Percent Agreement (NPA)	99.3% (559/563) (95%CI: 98.2% - 99.8%)
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Table 14: Cumulative PPA results by DPSO.

Days Since Symptom Onset	# Specimens Tested	# Cumulative Positive by Flowflex COVID-19 Antigen Home Test	# Cumulative Positive by RT-PCR	Cumulative PPA
0 to 1 day	203	47	53	88.7%
0 to 2 days	496	137	146	93.8%
0 to 3 days	643	169	184	91.9%
0 to 4 days	727	190	209	90.9%
0 to 5 days	774	204	227	89.9%
0 to 6 days	799	212	236	89.8%

C Clinical Cut-Off:

There is no clinical cut-off for this device. This section is therefore not applicable.

D Expected Values/Reference Range:

A patient sample is expected to be negative for SARS-CoV-2.

E Other Supportive Performance Characteristics Data:

1. Flex Studies

To assess the robustness of the Flowflex COVID-19 Antigen Home Test, flex studies were conducted that assessed all major aspects of the test procedure (sample volume, reading time, extraction buffer volume, bubbles in reagent tube, swab elution time and procedure, sample hold times before and during processing) and variability of environmental test conditions that the test may be subjected to when in use (Disturbance during run, Device orientation, lighting, various temperature and humidity stress conditions). Testing was performed with contrived positive nasal swabs generated by diluting heat inactivated SARS-CoV-2 virus into negative clinical nasal swab matrix at 2xLoD. The studies support that the test is robust in the intended use condition with an insignificant risk of erroneous result.

2. Usability Study:

A usability study was conducted to assess the lay user's ability to understand the instructions for use and to adequately execute the Flowflex COVID-19 Antigen Home Test workflow accordingly. A total of 431 subjects, ages 18 years and older, were enrolled in the study and were observed during testing. A total of 98.8% successfully completed testing by receiving a Negative or Positive result.

Following the usability study, all subjects were issued a questionnaire to assess users' comprehension of the test. The questionnaire was completed by 431 subjects enrolled in the EUA clinical study (one user missed one question). The questionnaire assessed users' understanding of concepts such as the test purpose and interpretation of results.

3. Readability Study:

The purpose of this study is to evaluate whether lay home-users (patients or their caregivers) can interpret test results correctly with low positive samples from the Flowflex COVID-19 Antigen Home Test.

The Readability Study was conducted according to an IRB approved protocol in a simulated home environment. A total of 61 lay users with diverse gender, ages and educational background who met the study inclusion criteria, were enrolled for the Readability Study. Each lay user was asked to interpret four test devices with three different concentrations that were arranged in blinded test panels with the following sample results; the order of the results within the panel was randomized:

- Negative, 1.5xLoD, 1.5xLoD, 5xLoD
- Negative, Negative, 1.5xLoD, 5xLoD

52% of the study participants were male and 48% of the study participants were female. 50% of individuals were vision impaired, and about 55% of individuals were either still at high school or had a high school degree as their highest level of education.

Table 15 Demographics - Age distribution and education of the clinical study cohort

Age group	Number of lay users	Percentage	Education	Lay user	Total
14-20 years	13	21%	Still at High School	9	15%
21-30 years	9	15%	High School or Equivalent	23	38%
31-55 years	25	41%	College	13	21%
> 55 years	14	23%	Bachelors	14	23%
			Masters	2	3%
			Doctorate	0	0%
Total	61	100%	Total	61	100%

*Tested by adult >14 years old per the intended use statement.

**24 users age 14 to <18 did not finish high school yet

Table 16: Vision impairment of lay user in readability study

Type of vision impairment	Yes	Percentage of vision impairment
Near sightedness only (with lens prescriptions)	17	28% (17/61 lay users)
Far sightedness only (with lens prescriptions)	8	13% (8/61 lay users)
Near sightedness and Far sightedness (with lens prescriptions)	5	8% (5/61 lay users)

Far sightedness and Glaucoma (with lens prescriptions)	1	2% (1/61 lay users)
Total lay users with vision impairment	31	51% (31/61 lay users)

Negative samples and samples with concentrations at 5xLoD were all correctly interpreted. Only samples with low concentration were occasionally interpreted incorrectly as negative. All of these false negative results were determined by the age group of >55 years of age, and were mostly (4/5) generated by individuals with vision impairment. It is therefore recommended that users with conditions affecting their vision- such as far-sightedness, glaucoma, or color blindness-are encouraged to seek assistance to interpret results accurately (e.g., reading glasses, additional light source, or another person).

Table 17: Readability study summary stratified by age and overall

Age group	Sample Concentrations			Correct/ Total
	Correct/ Negative	Correct/ Low Positive (1.5xLoD)	Correct/ Positive (5xLoD)	
14-20 years	22/22	17/17	13/13	52/52
21-30 years	13/13	14/14	9/9	36/36
31-55 years	36/36	39/39	25/25	100/100
> 55 years	21/21	16/21	14/14	51/56
Total	92/92	86/91	61/61	239/244
Negative Agreement	100% (92/92)	n/a	n/a	100% (92/92) (95%CI: 96%-100%)
Low Positive (1.5xLoD) Agreement	n/a	95% (86/91)	n/a	95% (86/91) (95%CI: 88%-98%)
Positive (5xLoD) Agreement	n/a	n/a	100% (61/61)	100% (61/61) (95% CI: 94%-100%)

IV Proposed Labeling:

The labeling supports the finding of substantial equivalence for this device.

V Conclusion:

The submitted information in this premarket notification is complete and supports a substantial equivalence decision.

FDA NEWS RELEASE

FDA Clears First COVID-19 Home Antigen Test

[More Press Announcements \(/news-events/newsroom/press-announcements/\)](/news-events/newsroom/press-announcements/)**For Immediate Release:**

November 09, 2023

[Español \(/news-events/press-announcements/la-fda-autoriza-la-primera-prueba-de-antigenos-covid-19-para-el-hogar\)](/news-events/press-announcements/la-fda-autoriza-la-primera-prueba-de-antigenos-covid-19-para-el-hogar)

Today, the U.S. Food and Drug Administration cleared for marketing the first over-the-counter (OTC) antigen test for COVID-19. ACON Laboratories' Flowflex COVID-19 Antigen Home Test, originally authorized for emergency use in 2021, is now the second home COVID-19 test to successfully complete a traditional FDA premarket review pathway, and the first indicated for use in children under 18. Today's announcement follows clearance of a molecular home test earlier this year.

"This marks the latest step forward in our efforts to help test developers provide Americans with continued options for safe and effective COVID tests that can be performed entirely at home," said Jeff Shuren, M.D., J.D., director of the FDA's Center for Devices and Radiological Health. "The FDA continues to proactively work with test developers that desire to market their products beyond emergency use authorities. This is part of the FDA's broader effort to advance the development and availability of at-home tests for a variety of medical conditions to expand patient access to testing."

The Flowflex COVID-19 Antigen Home Test is a visually-read test cleared for OTC home use by symptomatic individuals within six days of symptom onset. It is cleared for individuals aged 14 years or older testing themselves, or adults testing individuals aged two years or older. In a study reviewed by the FDA, this test correctly identified 89.8% of positive and 99.3% of negative samples in individuals with signs and symptoms of upper respiratory infection.

As with antigen tests authorized for emergency use, this test is intended to be used at least twice over three days with at least 48 hours between tests. This means that a symptomatic individual with an initial negative test result should be re-tested once between 48 and 72 hours after the first test using an antigen test for COVID-19 or followed up with a molecular COVID-19 test.

The FDA reviewed the ACON Flowflex COVID-19 Antigen Home Test through the 510(k) premarket review pathway. A 510(k) is a premarket submission made to the FDA to demonstrate that a new device is substantially equivalent to a legally marketed predicate device.

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The FDA, an agency within the U.S. Department of Health and Human Services, protects the public health by assuring the safety, effectiveness, and security of human and veterinary drugs, vaccines and other biological products for human use, and medical devices. The agency also is responsible for the safety and security of our nation's food supply, cosmetics, dietary supplements, products that give off electronic radiation, and for regulating tobacco products.

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